

INTRODUCTION

CAS is a neurological motor speech disorder affecting speech planning, requiring high-frequency, repetitive practice to support motor learning (Maas et al., 2014). By combining evidence-based therapy principles with game design, SPOLVE, a mobile game, aims to improve adherence to home practice, increase therapy dosage, and enhance clinical outcomes.

1. REQUIREMENTS

1.1. Stakeholders

Following Freeman (2010) and Clarkson (1998, pp. 258–261), stakeholders were classified as primary, those directly involved in SPOLVE's operation whose participation is essential to its function or secondary, those who influence or are affected by it without direct participation.

Stakeholder	Type	Role	Justification
Child (Aged 3-18)	Primary	Beneficiary: Primary User with CAS or related speech disorder. Uses the gamified interface for daily speech practice.	Children with special healthcare needs require intensive, high repetition intervention (Apkon & Myers, 2015), making engagement critical. Gamification helps overcome the monotony of traditional drills.
Speech-Language Pathologist (SLP)	Primary	Decision Maker/Expert: Clinician prescribing target words, managing lessons and reviewing session analytics.	SLP supports clinical decision-making, with session-level data enabling effective monitoring and therapy adjustment in line with structured assessment needs (Apkon & Myers, 2015, p.13).
Parent / Carer	Primary	Facilitator: Adult facilitator who manages the device and home practice sessions	Parents support consistent home therapy; SPOLVE reduces care and travel burden (Apkon & Myers, 2015) by embedding therapy into routines and using gamification to improve adherence and motor learning (Maas et al., 2014).
Regulatory Bodies (NHS / GDPR)	Secondary	Regulator: Authorities ensuring healthcare standards and child protection	Affects Objectives: Regulatory compliance is critical for handling sensitive speech data; measures like local storage and parental controls support adherence to the Data Protection Act 2018 (s.9) on parental consent for data processing.

1.2. Persona: Speech-Language Pathologist

Project adopts Persona methodology within the Goal-Directed Design framework established by Cooper et al. (2014), ensuring SPOLVE remains user-centred throughout the design process. A persona is defined as a “*composite archetype*” derived from user research that represents the goals, behaviours, and motivations of a specific user group. Designing for a clearly defined individual helps avoid the “*elastic user*” problem, where vague user assumptions lead to ineffective design decisions (Cooper et al., 2014).

Dr. Sarah Anne is selected as the primary persona because she is the clinical decision-maker. While the child directly interacts with the gamified interface, Sarah’s professional requirements for prescribing “dosage” and evaluating “Knowledge of Results” (KR) determine the system’s validity as a therapeutic tool. If the design fails to satisfy her need for data-driven control, SPOLVE risks being perceived as a recreational application rather than a legitimate clinical instrument.



Age: 34 | Cardiff, Wales

Workplace: NHS Paediatric Community Clinic

Qualification: MSc Speech-Language Pathology

Experience: 9 years

Technical Proficiency: Moderate-High

"In the clinic, I can guide their motor planning, but the real progress happens in the 167 hours a week they are not with me. I don't just need them to practice; I need to see the data from those trials so I can adjust their therapy plan effectively."

Overview

Dr. Sarah Anne is a paediatric Speech-Language Pathologist specialising in motor speech disorders, particularly Childhood Apraxia of Speech (CAS). She represents the primary persona, as her clinical requirements drive the system's design and functionality.

Background

Sarah manages a demanding caseload of 28 paediatric clients, eight diagnosed with CAS, six with phonological disorders, three with articulation disorders. CAS cases present the greatest challenge due to their underlying motor-planning deficits, requiring intensive and repetitive practice to develop accurate speech production. Sarah applies evidence-based interventions such as:

- DTTC(Dynamic Temporal and Tactile Cueing)
- ReST (Rapid Syllable Transition Training).

These approaches high-frequency practice (hundreds of repetitions per week) to be effective (Maas et al., 2014).

Currently, Sarah relies on paper-based word lists for home practice. However, parents often report that children become frustrated or bored within minutes, resulting in insufficient repetition ("low dosage") and a gap between prescribed therapy and actual outcomes.

Primary Goals

- **High Intensity Dosage(End Goal):** Deliver 100+ trials per home session to support motor learning and speech acquisition embedding repetition within engaging gameplay.
- **Data-Driven Decision Making(Experience Goal):** Access detailed performance data (e.g., per-word accuracy, transition errors) to adapt therapy strategies effectively.
- **Clinical Autonomy:** Create, modify, and assign custom word sets aligned with individual patient needs without technical barriers.
- **Knowledge of Results:** Provide immediate feedback to reinforce correct speech production and prevent error reinforcement.

Behaviours

- Conducts weekly therapy sessions and prescribes structured home practice
- Uses evidence-based frameworks but relies on manual tools (paper lists)
- Depends on parental feedback to assess adherence (often inconsistent)
- Adjusts therapy based on observed accuracy and progression

Frustrations (Pain Points)

- **Lack of Visibility into Home Practice:** No reliable data on whether sessions occur or if speech is produced accurately
- **Mismatch in Existing Tools:** Many applications focus on isolated sounds, rather than transitions required in CAS
- **High Parental Burden:** Parents feel like "drill sergeants" trying to get their children to practise, which strains the parent-child relationship

1.3. Functional Requirements

Derived by mapping stakeholder goals and the SLP persona into system behaviours. Requirements marked **MVP** represent the minimum viable set required to deliver therapeutic value.

ID	CATEGORY	REQUIREMENT	DESCRIPTION	JUSTIFICATION	MVP
F1	Profile	Profile Management	System shall allow creation, editing and deletion of player profile	Enables personalised therapy tracking while ensuring user control over data (GDPR)	✓
F2	Core Gameplay & Speech Interaction	Speech Based Interaction	System shall allow users to attack enemies through recognised speech input	Core therapeutic mechanism enabling speech practice through interaction	✓
F3		Target Word Display	System shall display lesson words during gameplay	Ensures structured and goal-oriented speech practice	✓
F4		Game Session Control	System shall provide configurable session duration (30–120s)	Prevents fatigue and supports controlled repetition cycles	✓
F5		Performance System	System shall provide clear performance tracking through lives and scoring.	Maintains engagement and focus through challenge and consequence	✓
F6		Gameplay Feedback	System shall provide real-time and post-session feedback	Supports Knowledge of Results (KR) and performance reflection.	✓
F7		Practice Mode Selection	System shall support different practice modes, including single-word, multiple-word, and combined sessions	Reflects established therapy structures for skill development	✓
F8		Difficulty Adjustment	System shall allow difficulty level selection	Supports autonomy and adaptive challenge.	✓
F9		Engagement Enhancements	System shall provide engaging feedback and visual variety to sustain user motivation during gameplay.	Maintains long-term engagement and reinforces user motivation through varied feedback	✓
F10	UI/UX & Accessibility	Core Interface Display	System shall display essential gameplay information (HUD, player status, rewards)	Ensures visibility of system status and supports engagement	✓
F11		Accessibility & Interface Settings	System shall allow adjustment of text size, light, contrast, layout, and orientation	Ensures accessibility across diverse user needs	✓
F12		Guided Interaction Support	System shall provide navigation hints and structured UI flow	Supports learnability for new/young users	✗
F13		Visual Engagement Design	System shall include themes, animations, and visual customisation	Maintains engagement in repetitive therapy tasks	✓
F14		Settings Interface	System shall provide a central interface for configuring system behaviour	Simplifies system control for non-technical users	✓

F15	Clinical Tools	Lesson Management	System shall allow creation, editing, activation and deletion of lessons	Enables personalised and adaptable therapy plans	✓
F16		Session & Performance Tracking	System shall record attempts, accuracy, and duration of sessions	Provides objective clinical data for analysis	✓
F17		Progress Monitoring	System shall visualise performance trends and word-level accuracy	Supports data-driven clinical decisions	✓
F18		Clinical Security	System shall restrict dashboard access with a recovery system	Protects sensitive clinical data	✓
F19		Word Bank Management	System shall allow management, search and filtering of words	Keeps therapy content relevant	✓
F20		Reporting Tools	System shall generate shareable reports	Supports communication between parents and clinicians	✗
F21	Gamification	Reward System	System shall provide rewards based on performance and enable their use within the system	Encourages repeated practice through positive reinforcement	✗
F22		Customisation System	System shall allow profile and gameplay customisation	Strengthens user attachment and motivation to gameplay	✗
F23		Competitive Features	System shall include leaderboard and ranking system with real-time updates	Introduces social motivation and sustained engagement	✗
F24	System	Application Control	System shall allow core navigation and safe exit	Ensures user control and usability	✓
F25		Device Permissions	System shall request for microphone access when required	Enables core speech recognition functionality	✓
F26		Brand Identity	System shall provide identifiable icon and name	Supports recognition and product credibility	✓

1.4. Non-Functional Requirements

ID	CATEGORY	REQUIREMENT	DESCRIPTION	JUSTIFICATION	MVP
NF1	Security	Local Data Storage	Data shall be stored locally where possible allow secure deletion when requested	Ensures GDPR compliance and privacy protection	✓
NF2	Performance	Recognition latency <100ms	Speech recognition shall occur in real-time	Maintains the “cause-and-effect” link critical in speech motor learning	✓
NF3	Accessibility	Icon-Based Interface	UI shall prioritise icons over text	Supports children with literacy or language delays	✓
NF4	Reliability	Session Auto-Save	System shall save data continuously	Prevents data loss and ensures reliability	✓
NF5	Focus	Cognitive Load Minimisation	UI shall be simple and organised	Allows focus on speech production tasks	✓
NF6	Safety	Parental Gate	Restricted areas shall require verification	Protects children from unintended access	✓
NF7	Availability	Offline Functionality	Core gameplay shall work without internet	Ensures accessibility in varied environments	✗

NF8	Stability	System Stability	System shall operate without crashes	Essential for trust in clinical applications	✓
NF9	Scalability	System Scalability	System shall support future expansion	Enables long-term growth and adoption	✓

1.5. Data Requirements

ID	CATEGORY	REQUIREMENT	DESCRIPTION	SOURCE	JUSTIFICATION	MVP
DR1	Speech Processing Data	Audio Input	Raw speech recordings	Device Microphone	Required for speech recognition	✓
DR2		Transcribed Speech	Text output from STT	Speech Recognition Engine	Enables comparison with target words	✓
DR3		Accuracy Scores	Calculated pronunciation scores	Internal Algorithm	Supports feedback and performance tracking	✓
DR4	Therapy and Gameplay Data	Target Word Sets	Words defined by SLP	SLP input	Core therapy content	✓
DR5		Session Logs	Attempts, successes, duration	System Logs	Enables dosage tracking	✓
DR6		Progress Data	Historical trends	Aggregated logs	Supports long-term clinical decisions	✗
DR7	User Data	User Profiles	Child identifiers(name, age)	Parent/SLP input	Enables personalised therapy	✗
DR8		Usage Data	Frequency and duration of sessions	System logs	Evaluates adherence to therapy	✓

2. PROTOTYPING

2.1. Overview

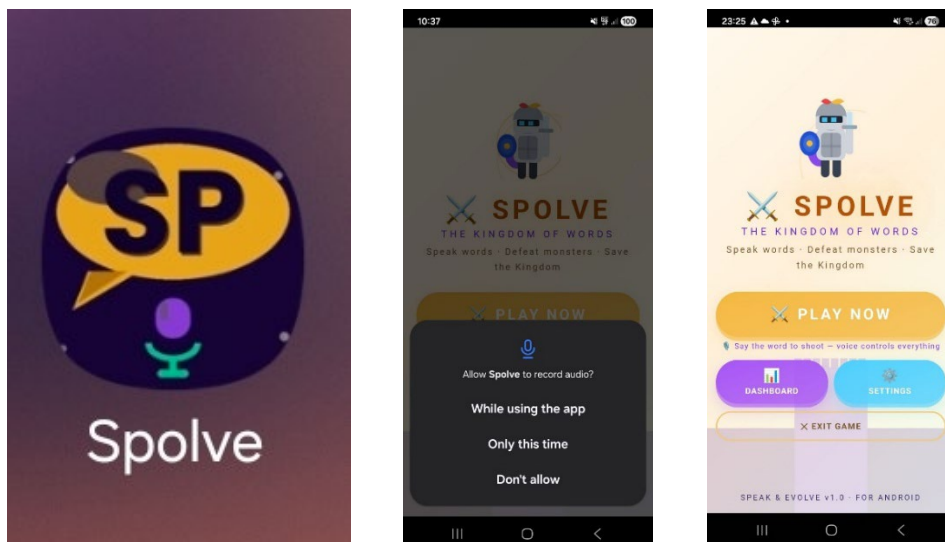
The SPOLVE ([app-debug.apk](#)) prototype supports both child users and Speech-Language Pathologists (SLPs) through a dual-interface structure. The system includes:

- A **main navigation interface** for accessing core features
- A **gameplay interface** for speech-based interaction
- A **settings interface** for configuration and accessibility
- A **clinical dashboard** for therapy management and progress tracking

SPOLVE supports direct interaction but lacks full system functionality; therefore, in line with Rudd et al. (1996), it is best classified as a medium-fidelity vertical prototype.

2.2. Screens and Requirement Mapping

2.2.1. Brand Identity & Device Permission



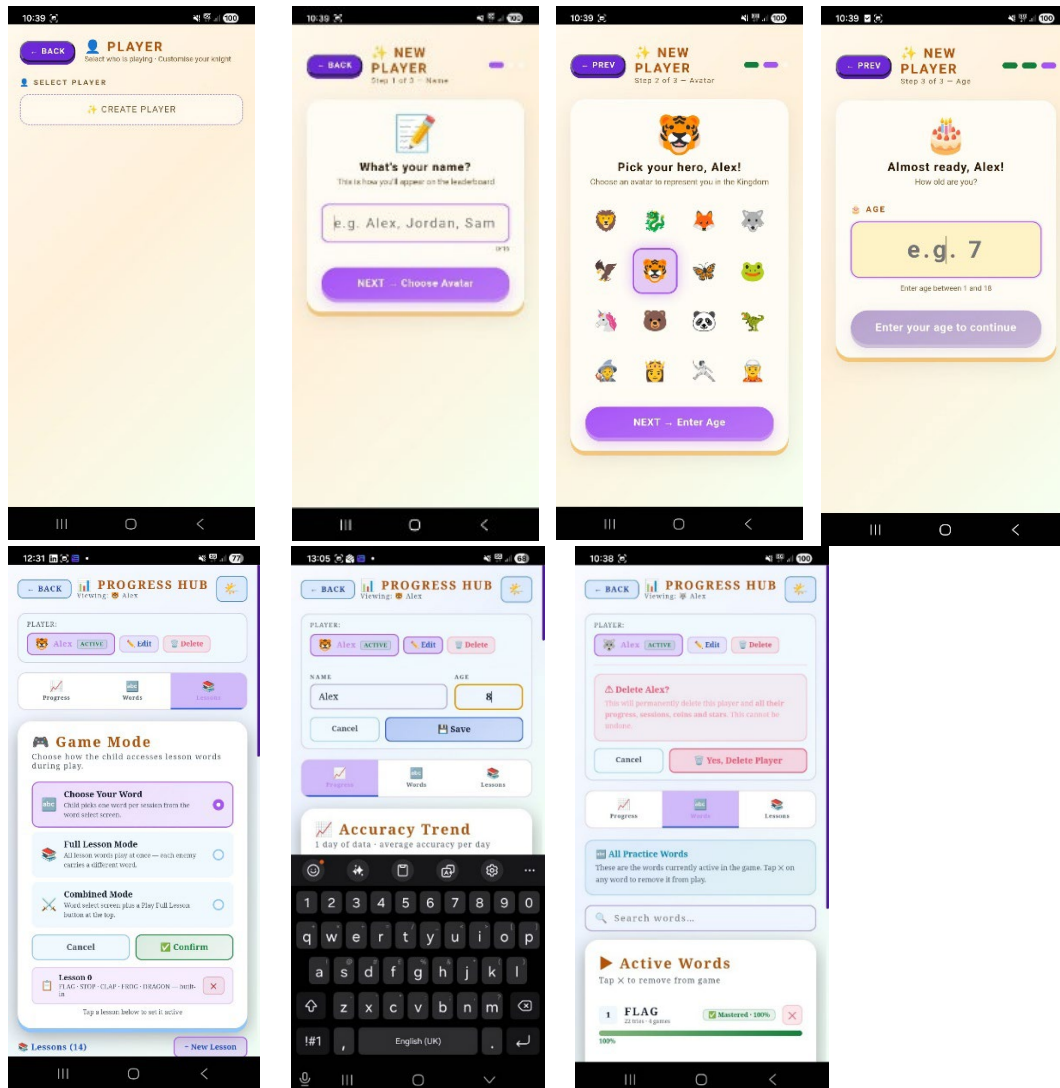
Interaction

Launch → grant mic access → navigate system ($F24 \cdot F25 \cdot F26$)

UX Justification

Featuring the Knight mascot and playful colours, the design uses the *Aesthetic-Usability Effect* and visual weight to guide attention toward key interactions (Lidwell, Holden and Butler, 2010). It prevents overload through *Hick's Law* with limited choices, optimises interaction via *Fitts's Law* with a large “PLAY NOW” button, and ensures accessibility by placing actions within the *Thumb Zone* to reduce strain (Lidwell, Holden and Butler, 2010; Hooper, 2013).

2.2.2. Profile Management



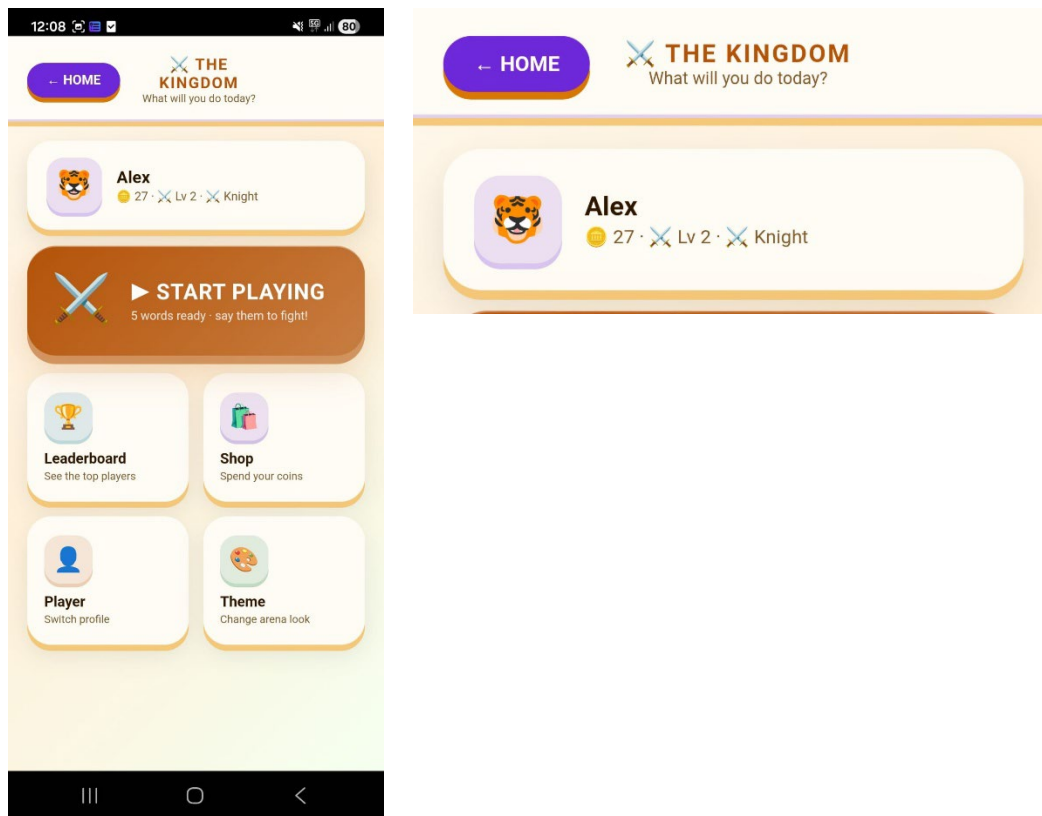
Interaction

Create → edit → delete (with confirmation)(F1 · F9 · NF1 · DR7)

UX Justification

Profile creation reduces performance load through *chunking* and *progressive disclosure*, with a progress bar supporting a clear mental model (Lidwell, Holden and Butler, 2010). Avatar selection uses the *picture superiority effect* and “hero” icons to aid literacy and immersion, while high-contrast “NEXT” and “BACK” buttons ensure intuitive mapping and *forgiveness* (Lidwell, Holden and Butler, 2010). *Layering* maintains a high *Signal-to-Noise Ratio*, and the “Delete Player” feature applies *constraint* with two-step confirmation to prevent errors, while edit mode ensures *consistency* by mirroring the original profile creation fields. (Lidwell, Holden and Butler, 2010).

2.2.3. Play Hub and Player Status



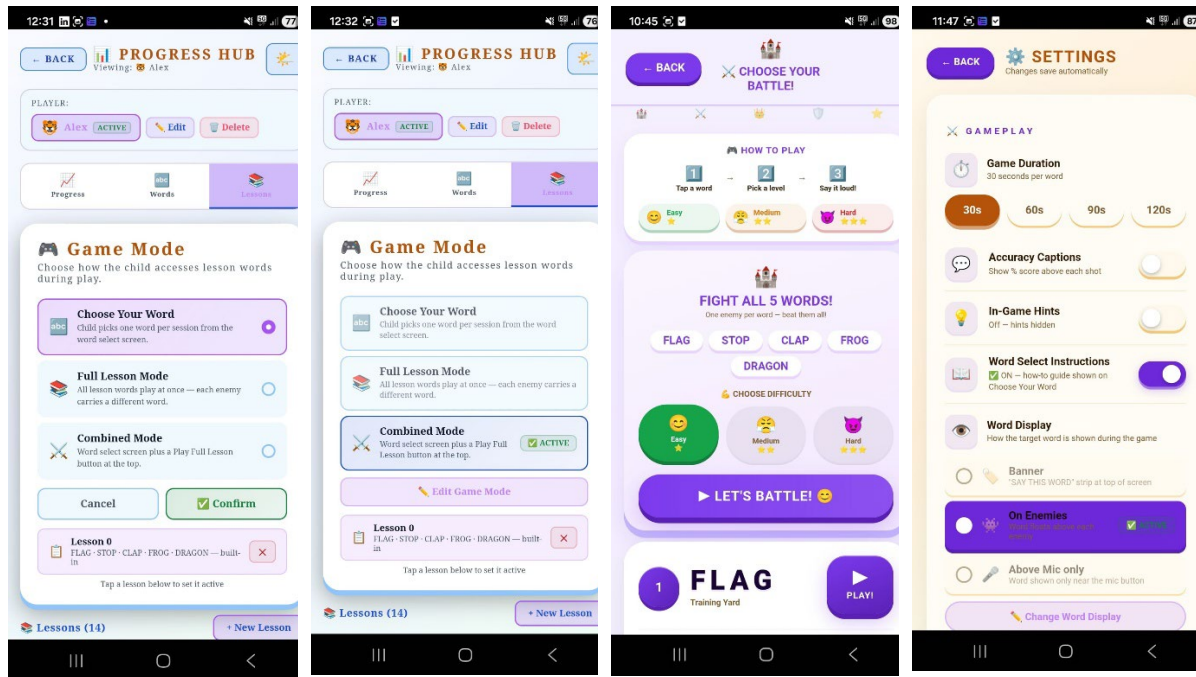
Interaction

View stats → navigate menus → start playing ($F10 \cdot F24 \cdot DR7$)

UX Justification

The persistent player card supports *Recognition Over Recall*, reinforcing the child's progress awareness (Lidwell, Holden and Butler, 2010). Its placement establishes a clear *Hierarchy*, with user status guiding navigation. Functions are *chunked into tiles* (Hick's Law) reducing decision time, while *Fitts's Law* prioritizes the "START PLAYING" button. *Icons* bypass literacy barriers and reduces performance load, while the "Home" anchor improves *wayfinding*, supporting proficiency (Lidwell, Holden and Butler, 2010).

2.2.4. Word and Difficulty Selection



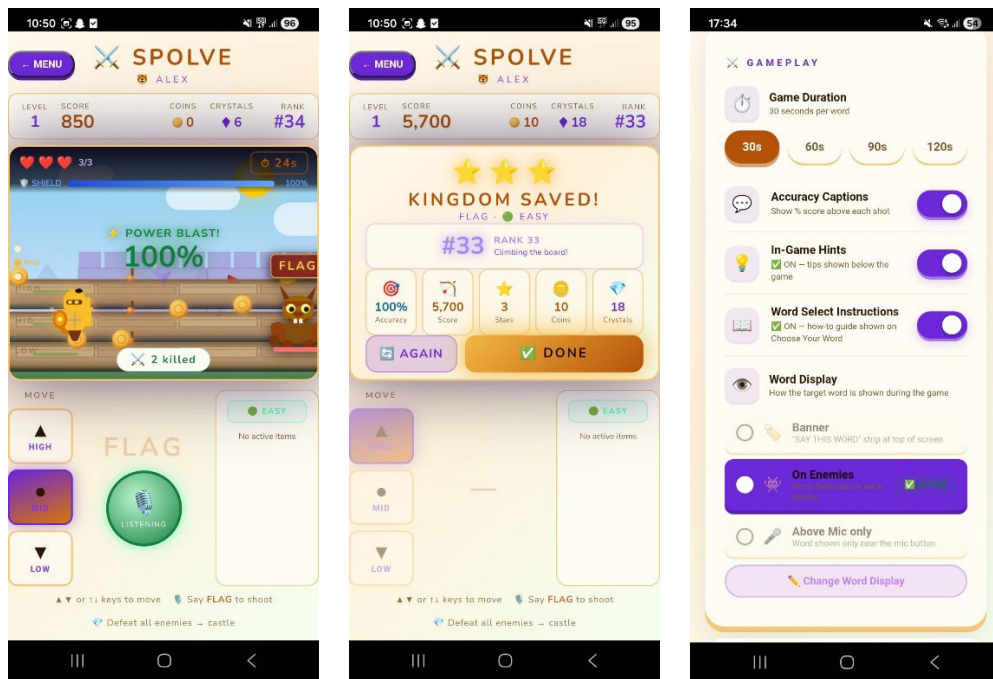
Interaction

- SLP : Edit mode → configure → confirm(F7)
- Child: Read instructions→ Select word → set difficulty → play→ adjust display(F8 · F12 · F14· DR4)

UX Justification

The feature balances clinical control and child autonomy using *Constraint* and an SLP-curated setup, managing the *Flexibility-Usability Tradeoff* so children avoid choice paralysis (Lidwell, Holden and Butler, 2010). *Segmenting* reduces cognitive load by separating configuration from word selection, while *Spatial Contiguity* and *Signaling* minimise extraneous processing (Mayer, 2020). *Colour* and *Highlighting* provide intuitive mapping and *iconic representations*, supported by personalization for engagement (Lidwell, Holden and Butler, 2010; Mayer, 2020). The “LET’S BATTLE!” button acts as a clear *Entry Point* in the *Terminal Area*, guiding users to action (Lidwell, Holden and Butler, 2010).

2.2.5. Core Gameplay & HUD



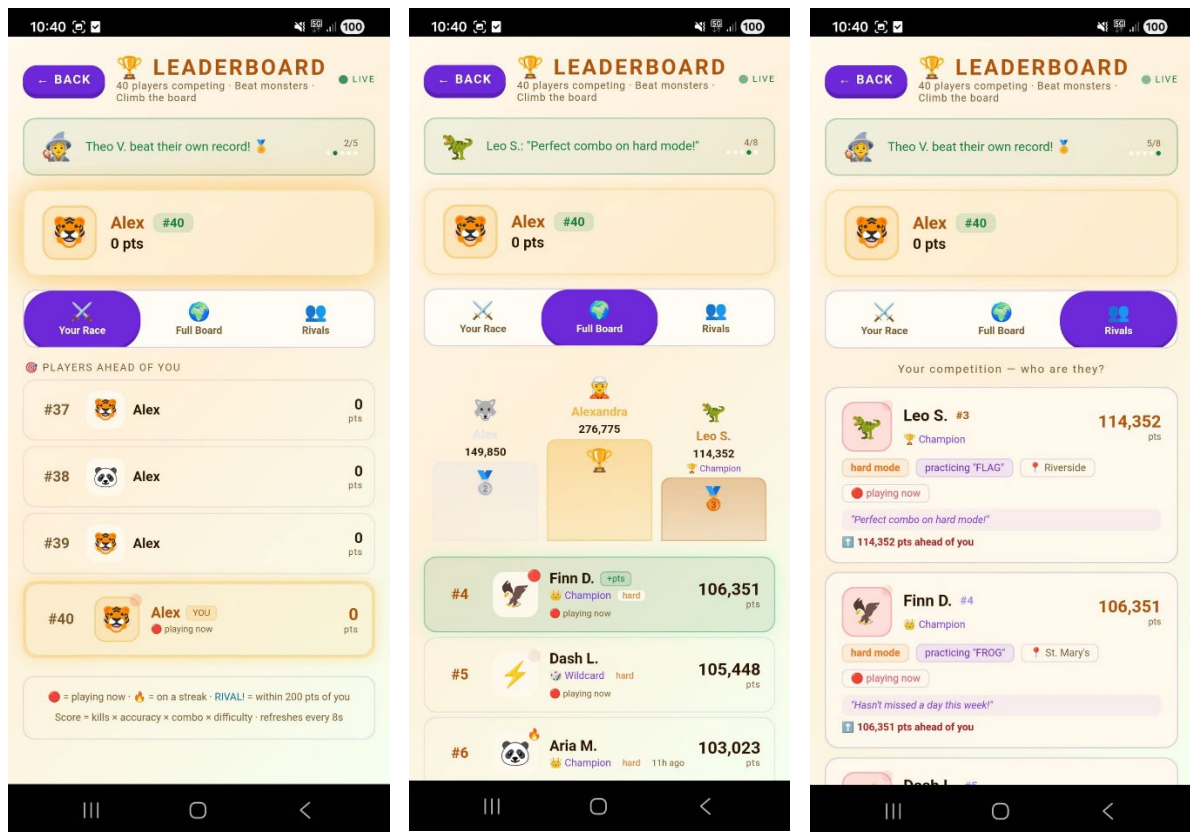
Interaction

Speak → navigate platforms → track via HUD → receive feedback → earn rewards → auto-save($F2 \cdot F3 \cdot F4 \cdot F5 \cdot F6 \cdot F9 \cdot F10 \cdot F12 \cdot F16 \cdot F21 \cdot NF2 \cdot NF3 \cdot NF4 \cdot NF5 \cdot NF7 \cdot NF8 \cdot DR1 \cdot DR2 \cdot DR3 \cdot DR4 \cdot DR5 \cdot DR8$)

UX Justification

The gameplay interface combines therapy and engagement by using speech-based interaction with real-time *feedback loops*, where accuracy captions and combat outcomes provide immediate knowledge of performance (Lidwell, Holden and Butler, 2010; Mayer, 2020). Controls apply *Mapping* and ergonomic placement to reduce *performance load*, aligning movement buttons with on-screen action (Lidwell, Holden and Butler, 2010; Hooper, 2013). The HUD supports essential processing through *Signaling* and *Spatial Contiguity*, while *iconic representations* (hearts, shields, crystals) improve accessibility for low-literacy users (Lidwell, Holden and Butler, 2010; Mayer, 2020). Gamification elements like rewards use *Operant Conditioning* to encourage practice, and auto-save ensures *forgiveness* and reliability, supporting user proficiency and sustained engagement (Lidwell, Holden and Butler, 2010).

2.2.6. Leaderboard System



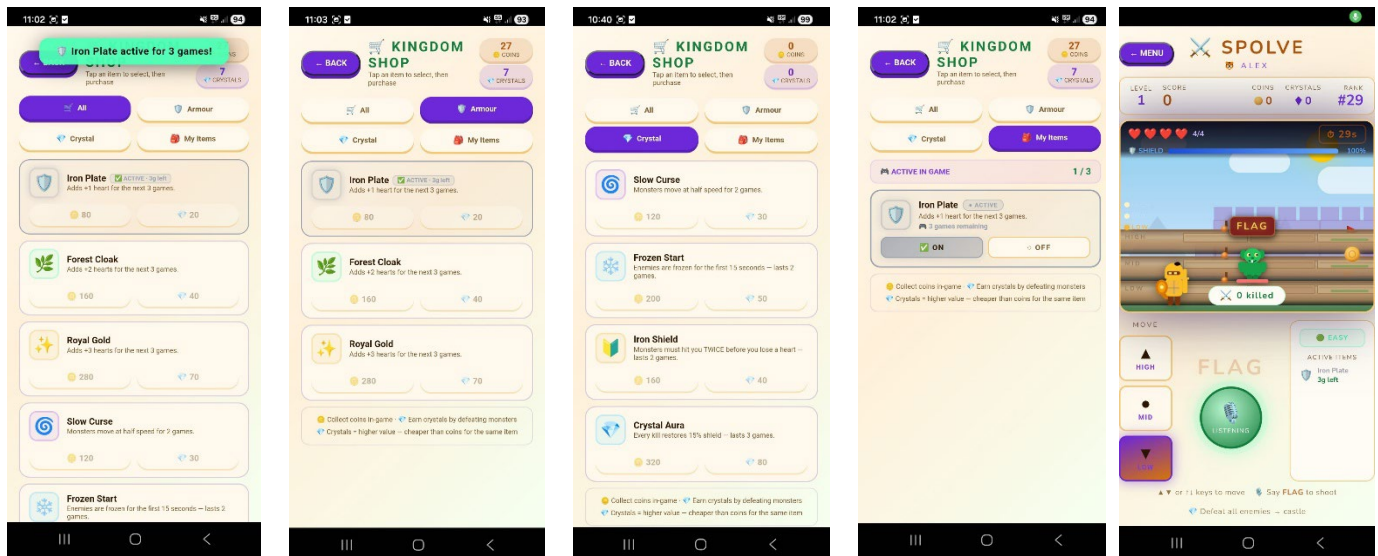
Interaction

View → switch tabs → track live rivals → rival data generated locally(F9· F23 · NF1· DR6· DR7)

UX Justification

The feature applies serious games principles to make clinical repetition socially engaging, using *iconic representation* to signal achievement (Ma et al., 2011; Lidwell, Holden and Butler, 2010). It counters the *N-effect* by using *segmenting* (e.g., “Your Race,” “Rivals”) to focus competition on small, relevant groups, maintaining motivation (Garcia and Tor, 2009; Mayer, 2020). Locally generated rival data ensures privacy, supporting *safety* and *trust* levels of the *Hierarchy of Needs* (Lidwell, Holden and Butler, 2010).

2.2.7. SHOP



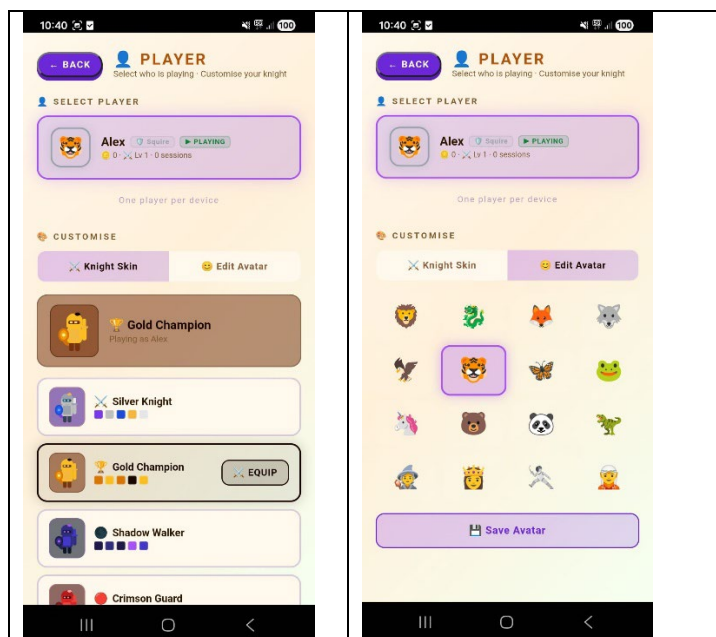
Interaction

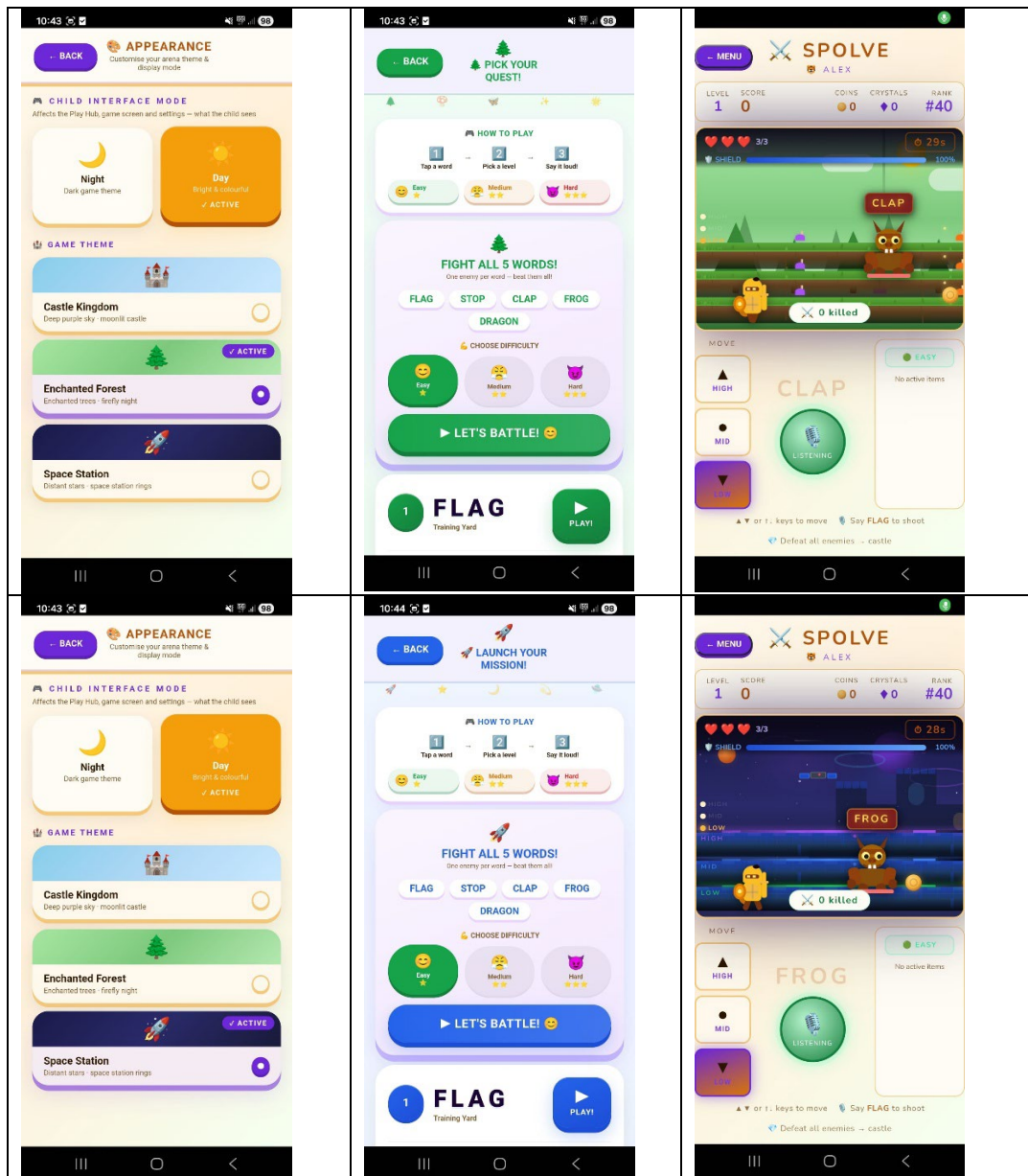
Browse → purchase → apply effects (HUD)(F9· F21)

UX Justification

Token economies enhance motivation by giving children control over how rewards are used, encouraging a shift from purely extrinsic rewards to more self-directed engagement (Deterding et al., 2011). A dual-currency system (coins and crystals) sustains short- and long-term engagement, maintaining *flow* (Csikszentmihalyi, 1990). The interface uses *Segmenting* to group items and prevent choice paralysis, while *iconic representation* reduces load for low-literacy users (Mayer, 2020; Lidwell, Holden and Butler, 2010). *Spatial Contiguity* ensures item effects are visible in the HUD, linking purchases to gameplay without extra scanning (Mayer, 2020).

2.2.8. Customisation and Theme System





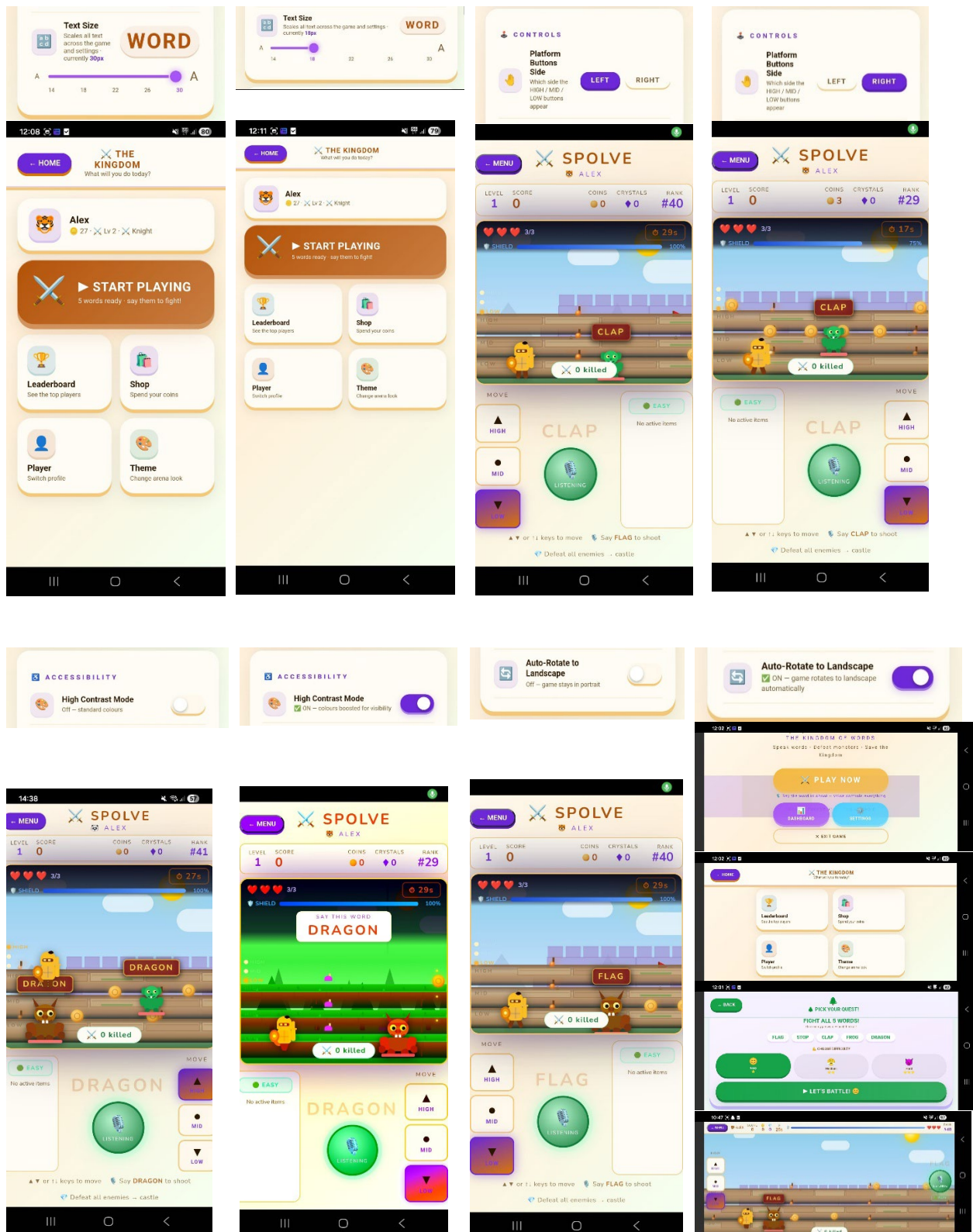
Interaction

- **Knight Skin:** Choose skin → preview → equip → applied in-game(F9· F22)
- **Avatar:** Select avatar→ confirm → updates globally(F9· F22)
- **Theme:** Select theme→ confirm → updates UI(F9· F13)

UX Justification

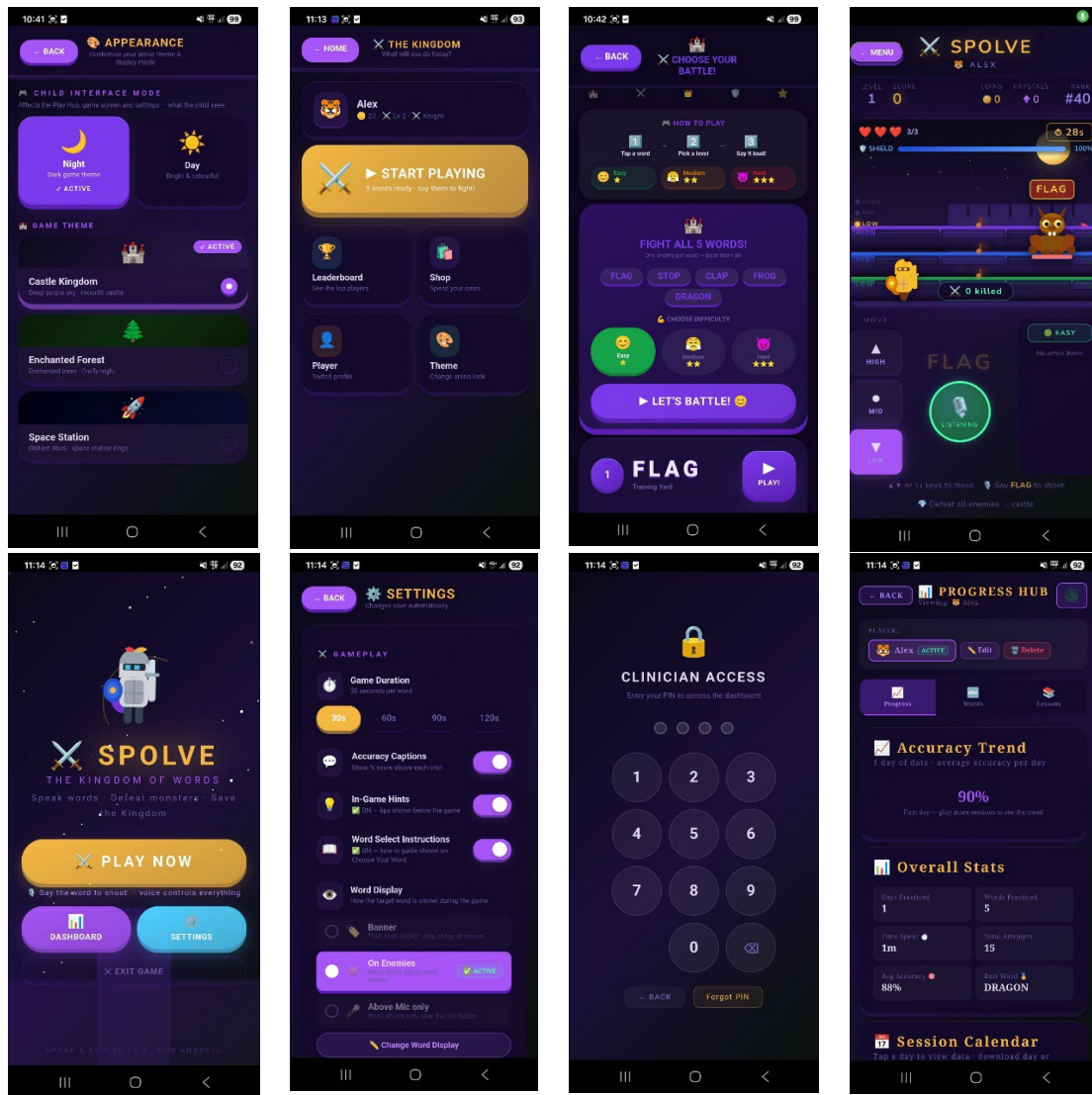
Player customisation supports long-term engagement by giving children agency and intrinsic choice over their environment (Deterring et al., 2011). Themes and avatars enhance motivation through variety and identification, fostering competence and participation (Prensky, 2003; Bowman, 1982). This aligns with *Aesthetic-Usability Effect* and supports *flow* (Csikszentmihalyi, 1990; Lidwell, Holden and Butler, 2010). A grid layout with *iconic representation* reduces cognitive load, while *highlighting* ensures clear system status (Lidwell, Holden and Butler, 2010; Mayer, 2020).

2.2.9. Accessibility & Interface Settings



Interaction

- Adjust text size, contrast, layout, orientation in the Settings interface(F11·NF3·NF5)



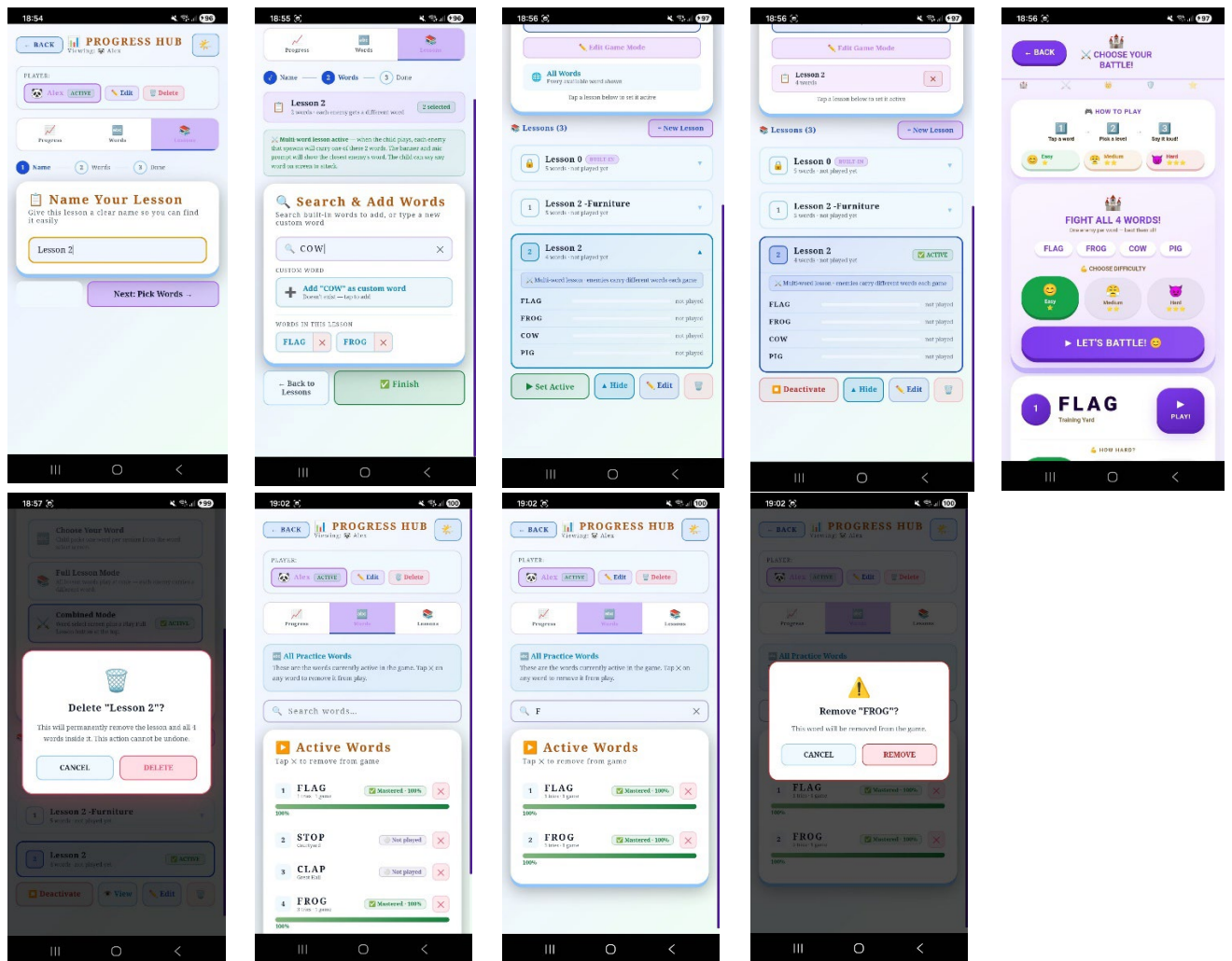
Interaction

- Child: Select theme → Select night → updates Play and Settings interface (F11 · NF3 · NF5)
- SLP: Select moon icon → updates dashboard interface (F11 · NF5)

UX Justification

The features enhance inclusivity through accessibility settings aligned with WCAG 2.1 and BDA guidelines, allowing adjustments to text size, contrast, and layout to support readability and neurodivergent users (W3C, 2018; BDA, 2018). A *Dark Mode* with independent toggles supports different users' needs in shared environments, aligning with SDG principles to reduce interference (Erickson et al., 2021; Shoemaker, 2001). Layout options accommodate handedness, applying *Fitts's Law* to improve interaction efficiency, while landscape orientation increases field of view and immersion, supporting flow (Papadatou-Pastou et al., 2020; Csikszentmihalyi, 1990; Lidwell, Holden and Butler, 2010).

2.2.10. Lesson & Word Bank Management



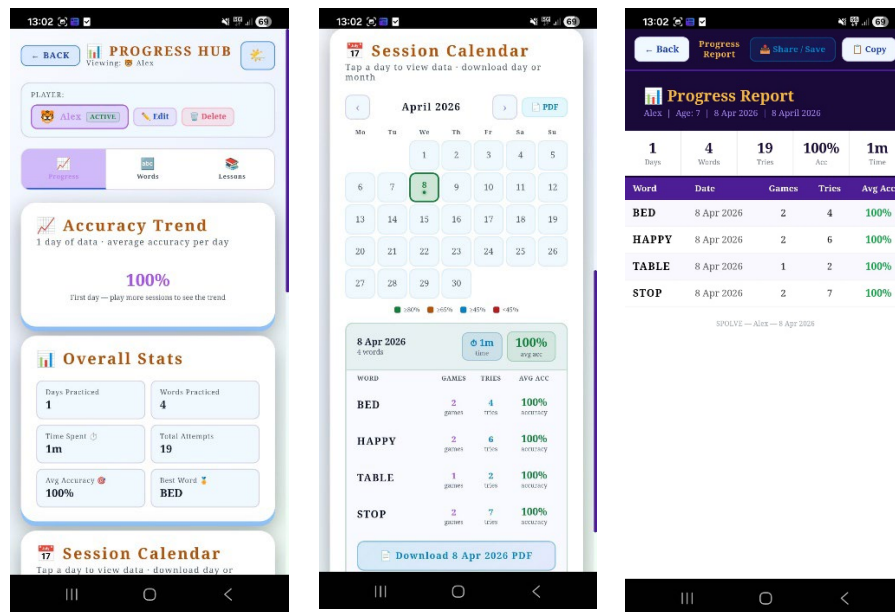
Interaction

- Lesson: Create → activate → appears in-game → edit → delete (confirm) (F15 · NF1 · DR4)
- Word: View → search → select → delete (confirm) (F19 · NF1)

UX Justification

A progress indicator supports the user's mental model through clear visibility of system status (Mayer, 2020). *Progressive disclosure* reduces cognitive load by separating tasks into stages, while confirmation dialogs apply *forgiveness* to prevent errors (Lidwell, Holden and Butler, 2010). The Word Bank uses *recognition over recall* and a *Gutenberg-based* layout for efficient decisions, with active/inactive toggles ensuring clear system status and flexible clinical use (Prensky, 2003; Lidwell, Holden and Butler, 2010; Mayer, 2020).

2.2.11. Session & Performance Tracking



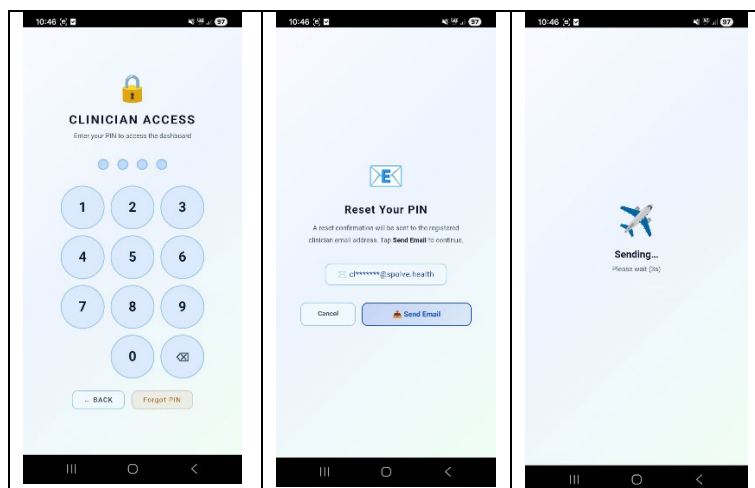
Interaction

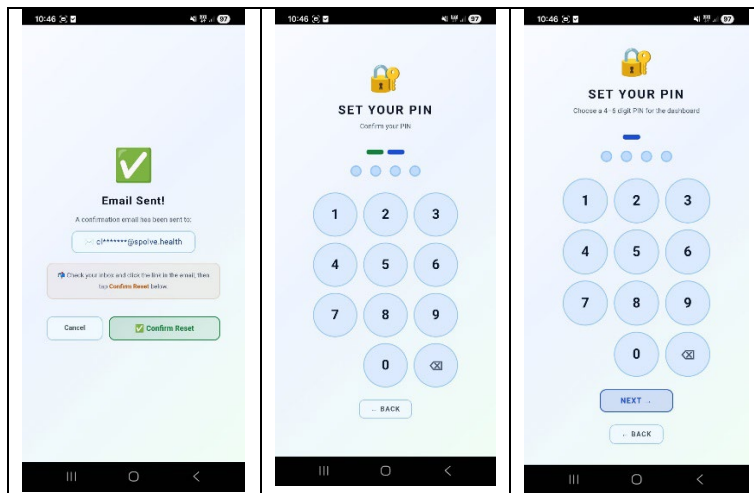
View → analyse → export(F17 · F20· DR3· DR5· DR6)

UX Justification

The dashboard reduces cognitive load through *progressive disclosure*, showing summaries before detailed metrics (Lidwell, Holden and Butler, 2010). It supports *external cognition* with visual data (e.g., accuracy, response time) to simplify pattern recognition (Mayer, 2020). Key export actions are placed in the *Gutenberg Diagram's terminal area*, enabling a seamless shift from analysis to documentation while maintaining clear system status of progress (Lidwell, Holden and Butler, 2010; Mayer, 2020).

2.2.12. Clinical Security





Interaction

Set → enter → reset PIN(F18)

UX Justification

The interface protects the therapeutic environment with a PIN-based *constraint*, preventing children from accessing administrative controls(Lidwell, Holden and Butler 2010). A clear keypad design reduces input errors, while email-based recovery balances security and usability by avoiding *task migration* and supporting *forgiveness*(Lidwell, Holden and Butler 2010). Isolating sensitive features also reduces extraneous processing, keeping the system focused and reliable during clinical sessions(Mayer 2020).

3. EVALUATION

The evaluation uses a three-phase triangulation approach: combining concurrent think-aloud (CTA), interviews, and questionnaires to reduce the gap between what users say and do. Mapping verbalizations to questionnaire items allows satisfaction scores to be validated against real-time user behaviour. This aligns with Nielsen (1993) and builds on Ichisani (2018), who show that CTA captures real-time experiences often missed in retrospective reports. Together, these methods enable a more accurate assessment by distinguishing true usability from responses influenced by surface aesthetics.

3.1. Questionnaires, Interviews and Observational Evaluation

3.1.1. Questionnaires

SPOLVE: PROTOTYPE EVALUATION QUESTIONNAIRE

Thank you for taking part in this evaluation. Please complete all questions based on your experience during the prototype interaction session. There are no right or wrong answers, your honest first impressions are exactly what we need.

This questionnaire should take approximately 10–15 minutes to complete.

kellydianga711@gmail.com [Switch account](#)

Not shared

* Indicates required question

SECTION A: USABILITY

Rating Scale: 1 = Strongly Disagree 2=Disagree 3= Neutral 4= Agree 5 = Strongly Agree

Q1. I found the SPOLVE interface easy to navigate. *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

Q2. I understood how to use the speech interaction (microphone/gameplay) without help. *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

Q3. The feedback (e.g. accuracy scores, rewards, animations) helped me understand my performance. *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

SECTION B: ENGAGEMENT & GAMIFICATION

Rating Scale: 1 = Strongly Disagree 2=Disagree 3= Neutral 4= Agree 5 = Strongly Agree

Q4. Which gamification elements do you think would most motivate a child to practise regularly?
(Tick up to 3 options only.)

- ☐ Enemy sprites and combat
- ☐ Leaderboard and rivals
- ☐ Coin and crystal rewards
- ☐ Knight skin customisation
- ☐ Game themes (Castle / Forest / Space)
- ☐ Shop and power-ups
- ☐ Accuracy feedback and star ratings
- ☐ Level progression

Q5. How engaging do you find the overall visual and audio design of SPOLVE? *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

Q6. I think a child would be motivated to use this system repeatedly over time *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

Q7. The difficulty levels (Easy/Medium/Hard) felt appropriate for learning progression. *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

SECTION C: CLINICAL VALUE & SYSTEM PERFORMANCE

Rating Scale: 1 = Strongly Disagree 2=Disagree 3= Neutral 4= Agree 5 = Strongly Agree

Q8. The system provides sufficient data (e.g., accuracy, attempts, progress trends) to support therapy decision-making. *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

Q9. I would feel confident using the Progress Hub to monitor a child's speech development over time. *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

Q10. The system allows easy creation and modification of therapy content (e.g., word sets, lessons). *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

Q11. I feel that the system provides appropriate security (e.g. password protection) for accessing sensitive clinical data

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

SECTION D: ACCESSIBILITY & DESIGN

Rating Scale: 1 = Strongly Disagree 2=Disagree 3= Neutral 4= Agree 5 = Strongly Agree

Q12. The interface was visually clear and accessible for children with different needs (e.g. e.g. dyslexia, visual impairment, left-handedness). *

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

Q13. The system provides sufficient options (e.g. text size, layout, contrast, guidance) to adapt the experience for different users.

1 2 3 4 5
☆ ☆ ☆ ☆ ☆

SECTION E: YOUR THOUGHTS

Please write your answers in the spaces provided.

Q14. What did you find confusing, difficult, or frustrating during your interaction with SPOLVE?
(Leave blank if nothing was confusing)

Your answer

Q15. What improvements would you suggest to improve usability or effectiveness of SPOLVE?

Your answer

Submit

Clear form

Link: [SPOLVE prototype evaluation questionnaire](#)

Section	Method	Rationale for Data Collection Mode	Expected data
A: Usability (Q1, Q2, Q3, Q7)	5-point Likert scale	Balances sensitivity and simplicity while reducing ambiguity and central tendency bias, with the agree/disagree format anchoring responses to meaningful positions and a five-point scale preferred over wider alternatives that offer no consistent reliability gains (Field, 2024; Likert, 1932; Clark et al., 2021).	Whether SPOLVE is intuitive, feedback is clear, and progression feels appropriate
B: Engagement and Gamification (Q4, Q5, Q6)	Q4: multi-select (max 3); Q5–Q6: 5-point Likert scale	Forcing prioritisation(Q4) yields genuine preference hierarchies over inflated ratings (Clark et al., 2021; Agile Business Consortium, 2014). Likert scales quantify engagement and retention perceptions in Q5 and Q6 (Likert, 1932).	Valued features, sustained engagement, and likelihood of repeated use
C: Clinical Value and Performance (Q8–Q11)	5-point Likert scale	Enables consistent comparison across clinical dimensions (Likert, 1932; Field, 2024)	Trust in dashboard, usability for clinicians, and adequacy of security (Cooper et al., 2014)
D: Accessibility and Design (Q12, Q13)	5-point Likert scale	Identifies gaps between features and real usability (Likert, 1932; W3C, 2018)	Whether accessibility works in practice and is sufficiently supported
E: Open-Ended (Q14, Q15)	Free-text responses	Captures issues beyond predefined measures and reduces social desirability bias (Norman, 2013; Clark et al., 2021)	Usability problems and suggested improvements, analysed thematically (Braun and Clarke, 2006)

3.1.2. Interview Protocol & Methodology

Methodology: Semi-structured Interview with **Think-Aloud Protocol**.

Think Aloud Tasks

#	Tasks	Mapping to Questionnaire
1	Create profile, navigate, play, adjust difficulty	Q1, Q2, Q3, Q7
2	View leaderboard and access shop	Q3, Q4, Q6
3	Switch dark/light mode, change theme, play	Q1, Q5
4	Switch to landscape mode and play	Q12
5	Adjust text size, word display, duration, play	Q13
6	Enable high contrast mode and play	Q12
7	Find session accuracy data	Q8, Q9, Q11
8	Access and share progress report	Q9
9	Create and activate lesson, enter game	Q1, Q10
10	Edit, modify, and delete lesson	Q10
11	Reset clinical interface password	Q11
12	Explore the rest of the application and additional features	Q14, Q15

Interview Questions

1. *During the tasks, I noticed you/observed hesitation or error]. Can you tell me more about what you were thinking in that moment?*

Reason:

It is a targeted *critical incident probe* (Flanagan, 1954) based on a specific event from Phase 2 think-aloud. Unique to each participant, it cannot be captured by questionnaires and turns observations into design insights by revealing root causes.

Expected data:

Root cause of a specific usability failure.

2. ***How would you describe SPOLVE to a parent of a child with CAS who had never seen a speech therapy app before? What would be your main concerns?***

Reason:

Externalises the user's mental model (Norman, 2013) by having them explain the system, revealing gaps between understanding and design intent, often the most insightful usability findings (Rubin and Chisnell, 2008). A parent-facing frame also exposes onboarding issues missed by questionnaires. This question adds qualitative context to Q1 and Q2 from the questionnaire by revealing what the participant believes they are interacting with.

Expected data:

- Conceptual Mapping: Game vs. clinical perception
- Salience: Key features identified
- Risk Elicitation: User anxieties and onboarding hurdles

3. ***Do you think any gamification elements (leaderboard, reward system), could be counterproductive for a child who is already anxious or self-conscious about their speech?***

Reason:

Requires nuanced reasoning beyond Likert scales, as features like leaderboards have dual effects, motivating some but discouraging others. Only dialogue can uncover when they become counterproductive, especially given potential frustration in children with speech disorders (Cummings, 2014; Clark et al., 2021). Unlike Q4, Q5 from the questionnaire that ask what is motivating, this asks what could be harmful.

Expected data:

- Feature Sensitivity: Competition-triggered anxiety
- Mitigation: Customisation or toggling options
- Self-Efficacy: Impact on confidence

4. ***If you could only keep three features of SPOLVE and had to remove everything else, which three would you keep and why?***

Reason:

A qualitative adaptation of *MoSCoW prioritisation* (Agile Business Consortium, 2014) that forces real-time justification, revealing deeper value systems behind choices rather than surface-level rankings. Links to Q4, Q10 but goes beyond ratings and forces trade-offs revealing essential vs nice-to-have features

Expected data:

- Feature Hierarchy: Essential therapy features
- Value Justification: Why features are prioritised
- Development Roadmap: Distinguishes must-haves from optional additions

5. ***Is there anything about SPOLVE that surprised you, positively or negatively, that you have not yet mentioned***

Reason:

A sensitive probe for unanticipated findings (Norman, 2013). Placed last so it acts as a qualitative safety net, capturing novel insights the previous questions and tasks did not elicit.

Expected data:

- Unanticipated Positives: Key “delighters” or added benefits
- Unanticipated Negatives: High-priority pain points
- Novel Use Cases: Unexpected applications or behaviours

3.2. Qualitative Evaluation

Reflexive thematic analysis (Braun & Clarke, 2006; 2019) was used to analyse collected qualitative data from think-aloud observations, interviews, and open-ended questionnaire responses (Q14, Q15). It was selected for its suitability in identifying meaningful patterns across multiple data sources within the same study and understanding users’ experiences and views (Caulfield, 2023).

An inductive approach (Caulfield, 2023) was applied, following Braun and Clarke's (2006) six-step process (familiarisation, coding, theme development, review, definition, and write-up). Reflexive TA also suited the researcher’s dual role as SPOLVE’s designer, treating perspective as legitimate part of the analytic process rather than a limitation.

Themes reflected shared patterns of meaning, not just topics. Triangulating think-aloud, interview, and questionnaire data helped validate findings and reduce bias (Jakob Nielsen, 1993; Ichسانی, 2018).

3.2.1. Coding

This was conducted line-by-line, assigning concise labels to capture behaviours and participant statements (e.g., hesitation, concern or confusion in think-aloud). In total, 34 inductive codes were generated across transcript T1(P1, Appendix B), transcript T2(P2, Appendix C) and questionnaire responses (R1, R2). Each code is tagged by participant, source, and timestamp/question.

ID	Code	Source
1	Unclear navigation path to core gameplay	P1 [T1: 00:55]
2	Information density overwhelming on leaderboard	P1 [T1: 05:25]
3	Interface assumes adult prior experience	P1 [T1: 06:16]
4	Self-blaming for recognition failure	P1 [T1: 11:08]
5	Landscape control layout causes friction	P1 [T1: 12:38-12:57]
6	Word displayed on enemy head aids comprehension	P1 [T1: 15:07-15:26]
7	High contrast improves spatial awareness in gameplay	P1 [T1: 16:12-16:30]
8	Progress data immediately useful to parent/clinician	P1 [T1: 18:15-18:22]
9	PDF export valued for clinical communication	P1 [T1: 18:32-18:45]
10	Feature richness positively surprising	P1 [T1: 25:24]
11	Gameplay eclipses speech engagement	P1 [T1: 26:24]
12	Light Mode visually overstimulating	P1 [T1: 26:57]
13	Gamification successfully masks therapy as play	P1 [T1: 29:01-29:30]
14	Therapeutic Risk: Avatar movement enables speech bypass	P1 [T1: 30:00-30:12]
15	Adversarial framing of Rivals triggers stress	P1 [T1: 31:13]
16	Social visibility acceptable; direct competition not	P1 [T1: 31:46]
17	Dashboard features (PDF, streak, lessons) prioritised as top-3 keeps	P1 [T1: 33:18-33:56]
18	Feature overload: requests compact mode for children	P1 [T1: 36:12-36:48]
19	Visual design positively resonates with child-centred intent	P1 [T1: 37:53-38:17]
20	Reward mechanic unclear to new user	P2 [T2: 02:02]
21	Latency at higher difficulty outpaces recognition	P2 [T2: 03:02]
22	Uncontrollable scrolling banner frustrates user	P2 [T2: 04:55]
23	Word Bank / Lesson data model mismatch	P2 [T2: 22:07]
24	Hard mode time pressure may reduce pronunciation care	P2 [T2: 27:06]

25	Latency creates mis-attribution of reward to wrong enemy	P2 [T2: 28:00]
26	False-positive reinforcement risks embedding mispronunciation	P2 [T2: 28:25]
27	Model under-sensitive to short/single-syllable words	P2 [T2: 29:18]
28	Global leaderboard scope demotivating for new users	P2 [T2: 29:59]
29	Dashboard enables targeted, data-driven therapy	P2 [T2: 30:54]
30	Movement controls add therapeutic engagement layer	P2 [T2: 31:25]
31	Landscape layout and small HUD elements reduce legibility	R3 [Q14]
32	PIN reset flow causes unnecessary disruption to clinical workflow	R3 [Q15]
33	Feature density confusing for children	R1 [Q15]
34	Initial onboarding friction which improves with familiarisation	R1 [Q14]

Note: Full excerpts are available in the highlighted transcripts (Appendices B and C). Each excerpt is colour-coded by theme: Speech Recognition (yellow), Cognitive/Visual Load (green), Gamification (pink), Clinical Utility (blue)

3.2.2. Questionnaire Results

SECTION E: YOUR THOUGHTS

Q14. What did you find confusing, difficult, or frustrating during your interaction with SPOLVE?
(Leave blank if nothing was confusing)

3 responses

It's a bit difficult at the very beginning of using the app. But when I settled in this app, I can use it without any confusing.

took me a while to realise hwo the game worked, some buttons could be simplified or made a bit clearer.

The landscape mode obfuscated a lot of elements and created more scrolling in menus. Sometimes there would be enemies spawned but time would expire. Some elements in game were quite small like the game timer. Some text elements overlapped in gameplay

Q15. What improvements would you suggest to improve usability or effectiveness of SPOLVE?

3 responses

Could you make this app more compact? If I were a kid, who has less experience in this kind of competition game, I would be a bit confused because of their a lot of functionalities.

maybe a quite tutuorial screen on first game, maybe make the microphone icon look less like a button

Adjust the sensitivity of menus to stop accidental presses when scrolling. Adjust the speed of the difficulties to be more consistent. Have a way to change the PIN without being logged out as a therapist

3.2.3. Collaborative Reflexivity

To enhance trustworthiness and reduce designer bias, a second analyst independently coded the transcripts without access to the original codebook (Braun and Clarke, 2019). This open coding approach ensured an independent interpretation free from the researcher's existing framing.

The second analyst produced 39 codes(Appendix C), followed by a collaborative comparison. Differences were treated as opportunities for reflexive dialogue rather than errors, enriching interpretation rather than enforcing consensus.

Three outcomes emerged:

1. **Conceptual matches** strengthened confidence in themes - both analysts independently identified leaderboard information overload [P1: T1: 05:25].

2. **Partial matches** refined interpretations by combining both perspectives - gameplay was coded as both bypassing speech input and distracting from pronunciation focus, sharpening Theme 3's central organising concept.
3. **Unique codes** from the second analyst, including accessibility settings warping the layout [T2: 11:57] and a performance slowdown creating confusion [T2: 07:24], were incorporated where they added depth, highlighting areas the primary analyst may have overlooked due to designer familiarity.

3.2.4. Theme Development, Review and Definition

Codes were iteratively refined into four themes (Braun & Clarke, 2006; 2019): speech recognition, cognitive/visual load, gamification, and clinical value.

ID	Theme	Codes	Definition	Review Decision
1	Speech Recognition	4, 21, 24, 25, 26, 27,36	Latency and inaccuracy disrupt feedback, risking reinforcement of mispronunciation and weakening therapy effectiveness.	Code 4 (self-blame) retained as clinically relevant in a child population. Code 36(T2-S1-06) added from collaborative review
2	Cognitive/Visual Load	1, 2, 3, 5, 7, 12, 18, 20, 22, 31, 33, 34,35,37,38,39	Excessive on-screen information overwhelms users. Accessibility settings exist but can disrupt layout rather than resolve the issue	Code 34 retained to mark onboarding difficulty. Codes 35(T2-S1-09), 37(T2-S1-10), 38(T2-S1-08), 39(T2-S1-01) added from collaborative review
3	Gamification	11, 13, 14, 15, 16, 28	Competitive features feel adversarial while game mechanics allow therapeutic bypass risking motivation and therapeutic progress	No new codes added, second analyst's gamification codes mapped onto existing codes.
4	Clinical Value	6, 8, 9, 10, 17, 19, 23, 29, 30, 32	Dashboard features (lessons, progress, reports) define SPOLVE as a therapy tool rather than a game.	Workflow issues (code 23) moved here. No new codes added from collaborative review.

Note: Codes 35,36,37,38 and 39 are obtained from Appendix C

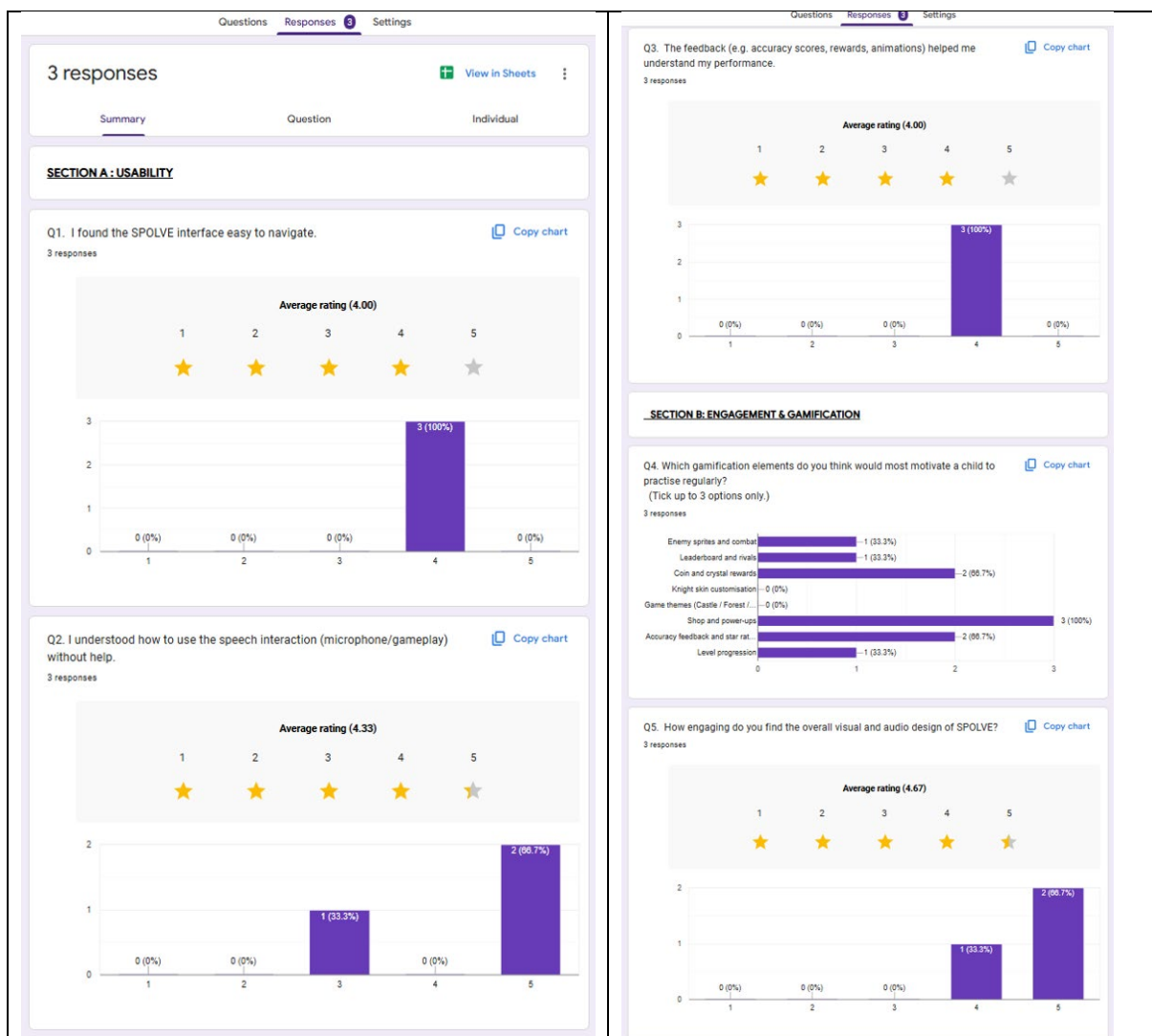
3.2.5. Improvements

Theme	Issue	Improvement	Affects
1	Recognition reliability	▪ Integrate a CAS-tuned speech recognition model with more tolerant phoneme matching for short words. ▪ Add a small processing buffer so delayed inputs are still correctly attributed ▪ Include a real-time listening indicator to clarify when input is being captured.	F2, F6, NF2
2	Cognitive/visual load	▪ Add a guided tutorial for new users to explain gameplay, controls, and HUD elements, addressing onboarding confusion ▪ Simplify leaderboard by defaulting to “Your Race,” with other views hidden behind an extra tap to reduce overwhelm. ▪ Improve HUD readability by enlarging key elements(e.g., timer) ▪ Fix layout issues in landscape mode to prevent overlap or cropping	F10, F11, F13, NF3, NF5
3	Gamification risk	▪ Reframe competitive elements (e.g., “Rivals”) to reduce pressure.	F21, F23, F5, F16

		Introduce safeguards such as a minimum speech-attempt threshold so gameplay cannot progress without meaningful speech input, preserving therapeutic intent.	
4	Clinical workflow integrity	- Improve system consistency by linking Word Bank and Lesson data so custom entries are reusable and searchable. - Refine the PIN reset flow to prevent workflow disruption, ensuring therapists can manage sessions without interruption.	F15, F18, F19

3.3. Quantitative Evaluation

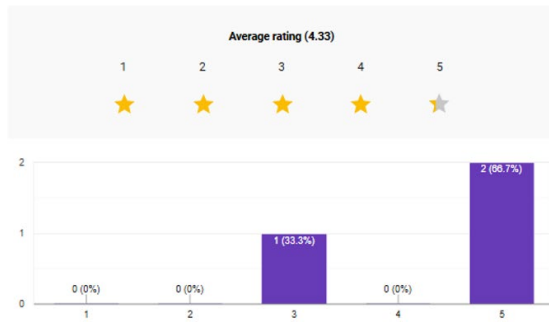
3.3.1. Results from the Questionnaire



Q6. I think a child would be motivated to use this system repeatedly over time

[Copy chart](#)

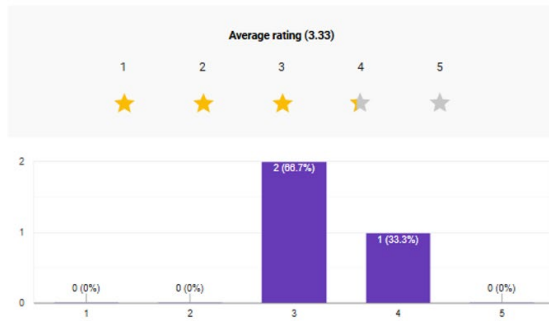
3 responses



Q7. The difficulty levels (Easy/Medium/Hard) felt appropriate for learning progression.

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3 responses

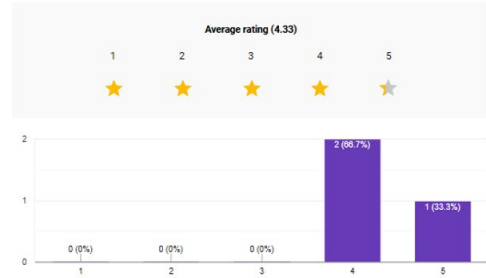


SECTION C: CLINICAL VALUE & SYSTEM PERFORMANCE

Q8. The system provides sufficient data (e.g., accuracy, attempts, progress trends) to support therapy decision-making.

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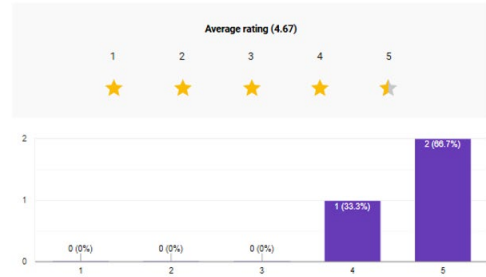
3 responses

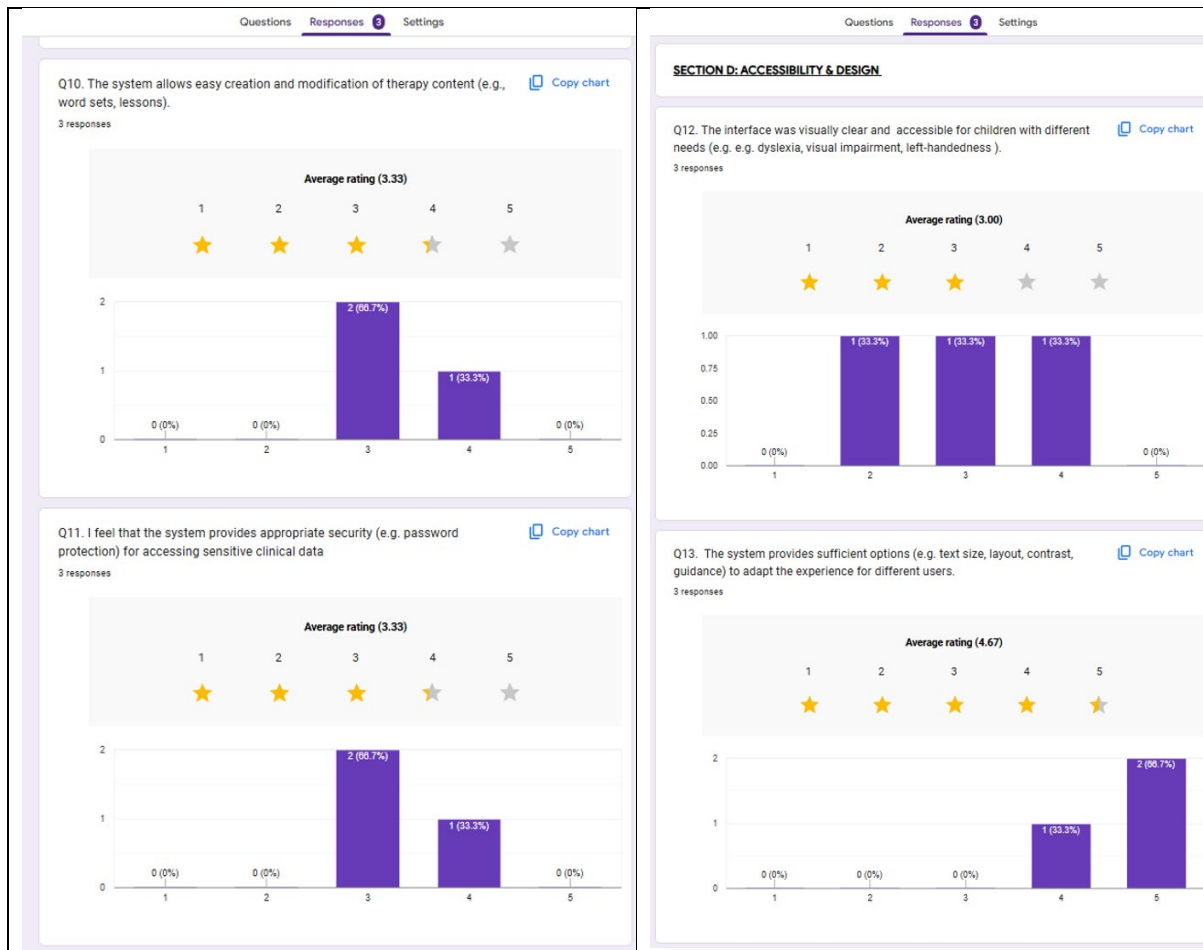


Q9. I would feel confident using the Progress Hub to monitor a child's speech development over time.

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3 responses





3.3.2. Descriptive Statistics

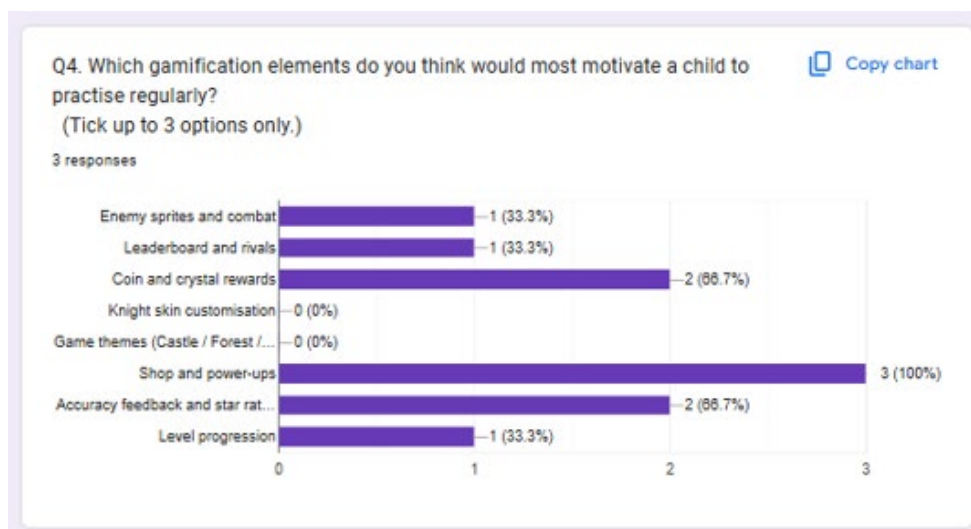
The table below presents descriptive statistics for all thirteen Likert items: mean (average rating), mode (most common response), Standard Deviation (SD) (participant agreement: low = strong, high = weak), range (spread between the highest and lowest individual responses) (Field, 2024).

Construct	Item	Description	Mean $= \frac{\sum x_i}{n}$	Median (middle value after sorting)	Mode (most frequent value)	SD $= \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$	Range $\max(x) - \min(x)$	Interpretation
Usability	Q1	Ease of navigation	4.00	4.0	4	0.00	0	Uniform agreement that navigation is easy
	Q2	Speech interaction understood	4.33	5.0	5	1.15	2	Understanding varies across users
	Q3	Feedback helped understanding	4.00	4.0	4	0.00	0	Feedback consistently perceived as helpful
	Q5	Engaging visual/audio design	4.67	5.0	5	0.58	1	Consistently high engagement

	Q7	Difficulty levels appropriate	3.33	3.0	3	0.58	1	Lowest usability rating, weaker performance
Retention	Q6	Child motivated to use repeatedly	4.33	5.0	5	1.15	2	Motivation varies across users
Clinical	Q8	Sufficient therapy data	4.33	4.0	4	0.58	1	Data consistently seen as sufficient
	Q9	Confident using Progress Hub	4.67	5.0	5	0.58	1	High confidence levels across users
	Q10	Easy lesson/word set creation	3.33	3.0	3	0.58	1	Consistent and moderate ease of use
	Q11	Appropriate security for clinical data	3.33	3.0	3	0.58	1	Security seen as adequate
Accessibility	Q12	Visually clear/accessible	3.00	3.0	2	1.00	2	Inconsistent accessibility performance
	Q13	Sufficient adaptation options	4.67	5.0	5	0.58	1	Options consistently recognised

3.3.3. Frequency Analysis

Frequency analysis was selected for Q4 because it identifies which options were most and least preferred across participants by counting selections and expressing them as percentages, making patterns in categorical data immediately visible (Field, 2024).



Shop and power-ups (100%) indicate reward mechanics are the main driver of engagement. Preference for rewards and feedback (66.7%) over competition suggests motivation is driven more by personal progress than comparison. Low leaderboard selection (33.3%) implies competition should be secondary, while no interest in cosmetic features suggests limited impact on motivation.

3.3.4. Inferential Statistical tests

Assuming a larger sample, inferential analysis would determine whether observed differences are statistically significant. Given the ordinal nature of Likert data and the likelihood of non-normal distribution, non-parametric tests are prioritised as they do not assume normality and are robust for ranked data (Freeman and Campbell, 2006).

Test	Applied To	Reason for Test	If $p < 0.05$
Wilcoxon Signed Rank	Q1(navigation ease): same participants pre/post tutorial	Evaluates whether a tutorial resolves onboarding friction (Theme 2). Uses paired ordinal data , ranking pre/post differences without assuming normality, making it appropriate for Likert responses (Webb, 2024; Freeman & Campbell, 2006).	Onboarding significantly improves usability and validating recommendation from Theme 2
Mann-Whitney U	Q8-Q11 (Clinical Utility): SLP vs non-SLP	Is the gap driven by SLP users finding content management inadequate. Compares two independent groups on ordinal data without assuming normality or equal variance (Freeman and Campbell, 2006).	Lower Q10–Q11 ratings among SLPs confirm a clinician-specific usability gap in content management

3.3.5. Suggested Improvements

Speech Interaction & Motivation Variability (Q2, Q6 — SD = 1.15)

Improve speech interaction feedback (e.g., clearer prompts, timing flexibility) and reinforce motivation through reward-based systems, prioritising rewards over competitive features.

Gamification Design (Q4 Frequency Findings)

Emphasise reward-based progression (coins, power-ups, feedback) while positioning competitive elements as optional.

3.4. Heuristic Accessibility Evaluation

WCAG 2.1 Level AA (W3C, 2018) guided the evaluation, using the POUR principles (Perceivable, Operable, Understandable, Robust) with evidence drawn from qualitative themes (T1–T4), initial codes, and transcript timestamps.

WCAG Criterion	Current State VS Gap	Recommended Fix
WCAG 1.1.1 Non-text Content	Pass: Icons are generally paired with text (e.g., navigation, HUD) Gap: standalone icons (hearts, shield, coins) and emoji difficulty cues may not be accessible to screen readers.	Add accessible names to all icons and provide text equivalents for emoji indicators.
WCAG 1.3.3 Sensory Characteristics	Pass: Has multi-modal feedback (visual, captions, haptics) Gap: correct/incorrect recognition relies mainly on visuals, limiting accessibility for low-vision users.	Add clear audio cues for correct/incorrect recognition and victory/defeat, independent of visuals.
WCAG 1.4.1 Use of Colour	Pass: Information is consistently presented in multiple ways. Colour is supplemented by shape and text (hearts for lives, difficulty icons, and distinct enemy sprites). Meaning is never conveyed by colour alone.	Maintain multi-modal signalling for all future UI elements
WCAG 1.4.3 Contrast(Minimum)	Pass: A High Contrast mode is available via Settings.	keep High Contrast mode

WCAG 1.4.4 Resize Text	Pass: Adjustable text size is available via Word Display settings. Gap: Enlarged text can overlap HUD elements during gameplay.	Implement responsive UI containers to prevent menu warping; ensure layout adapts dynamically to text scaling.
WCAG 1.4.8 Visual Presentation	Pass: Provides a mechanism to toggle between Light and Dark modes. Gap: Text scaling causes menu warping and layout disorientation.	Fix CSS/Layout containers to allow for responsive resizing without warping when text is enlarged.
WCAG 1.4.11 Non-text Contrast	Pass: Clear affordance; users easily identify "disabled" shop states. Gap: Visual distinction relies on a "faded" effect (reduced opacity) which may not meet the 3:1 contrast ratio required for UI components.	Implement distinct visual markers for disabled states (e.g., a "locked" icon or specific pattern) rather than relying solely on opacity.
WCAG 2.1.1 Keyboard	Pass: Tap targets and adjustable layouts (e.g., left-handed mode) support varied physical reach. Gap: Interaction is touch-only; no support for external keyboards or switch-access for users with severe motor impairments.	Implement full keyboard mapping for menu navigation to enable switch-interface compatibility.
WCAG 2.2.1 Timing Adjustable	Pass: Session durations are configurable. Gap: Per-word recognition windows are fixed; clinicians cannot adjust individual response timing for children with slower processing speeds.	Implement a response-window multiplier in the clinician dashboard to allow customized per-word timing based on the child's specific profile.
WCAG 2.5.4 Motion Actuation	Pass: Manual orientation settings (Portrait/Landscape) are available in the menu. Gap: Lack of a mid-session orientation lock allows auto-rotation to reorder controls and crop elements, disorienting the user.	Implement a session-specific orientation lock toggle and disable auto-reflow during active gameplay to maintain UI consistency.
WCAG 4.1.2 Name, Role, Value	Pass: Secure PIN-based access includes email recovery (supporting the principle of forgiveness). All administrative controls are protected by a verification layer to prevent unauthorized access.	Ensure every button and box is labelled in the code so screen readers can explain them.

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APPENDIX A- Interview Transcript 1

Participant 1 = Speaker 2

Speaker 1 (00:01): All right, so this is the start of the interview. And we are going to start to think aloud protocol.

Speaker 2 (00:07): OK

Speaker 1 (00:08): So what I need for you to do is to go to play now and create a player profile.

Speaker 2 (00:16): Create a profile.

Speaker 1 (00:18): It's been hold.

Speaker 2 (00:19): I play now okay.

Speaker 1 (00:21): Then, create a player profile for me.

Speaker 2 (00:24): Choose a maybe this one

Speaker 2 (00:32): It's like broke. How old, am I?

Speaker 1 (00:42): You can put any age that you want.

Speaker 2 (00:45): Yeah

Speaker 1 (00:50): All right, so I need you to navigate and play a game

Speaker 2 (00:55): Play a game where is play the game.

Speaker 2 (00:57): Happiness, customizing.

Speaker 2 (00:59): Ah so this this fragment is for customizing my Avatar, okay. High lip gluon, oh yeah. Hmm, okay.

Speaker 2 (01:14): and then, how can I play the game this button? In my profile. Oh, okay.

Speaker 1 (01:28): Play a game so when you play it. I need you to adjust the difficulty mode.

Speaker 2 (01:35): Ah, okay? So.

Speaker 1 (01:40): So I just pick any and play

Speaker 2 (01:43): Yeah What is? So the first section is just for all the process, and then the below thing is I can choose the ones that will face. Oh, yeah, okay.

Speaker 1 (02:06): feel free to just play it.

Speaker 2 (02:12): Clap cla.

Speaker 2 (02:14): Frog. frog, stop.

Speaker 2 (02:21): Oh

Speaker 1 (02:25): You need to have to say them all.

Speaker 2 (02:27): I just tried to even if I could not speak anything clap. Clap, flag, ooh oop. Okay.

Speaker 2 (02:40): Okay.

Speaker 1 (02:44): Play yea play again.

Speaker 2 (02:49): Oh I see, so this word is the which one I have to say it. Okay, stop. Start.

Speaker 2 (03:00): Play Ah

Speaker 2 (03:04): Frog. Frog.

Speaker 2 (03:14): Hey Stop. Stop, stop, stop. Hi, oh okay.

Speaker 2 (03:28): Oh okay, okay done again.

Speaker 2 (03:41): Yeah

Speaker 1 (03:42): Okay, just play again, so just pick any word.

Speaker 2 (03:47): Ok, ok. Okay, I'm gonna pick yeah dragon and then hard mode. Oh, yeah, this looks really dragon.

Speaker 2 (04:01): Dragon. Dragon. Dragon.

Speaker 2 (04:09): Ok.

Speaker 2 (04:12): Dragon. Dragon. My mom, why am I shield is gone?

Speaker 2 (04:22): Dragon.

Speaker 1 (04:25): All right, good, alright. Um, I need you to go to the leaderboard.

Speaker 1 (04:31): Hmm

Speaker 2 (04:34): done, yeah.

Speaker 1 (04:35): Just go to the leaderboard, see your ranking and just explore that interface. And just tell me what you think. As you explore?

Speaker 2 (04:48): I'm taking me some second to understanding your fragment. Oh, okay. So in your leader board, you have some looks like a banner top of them and then my profile and I can get reach in my race and full board and the other Candidates, peoples.

Speaker 2 (05:25): Okay, okay, I guess yeah, for oh can I speak in my opinion okay for me at first? The impression is, oh, **there are a lot of information in this one fragment. so that's why I say, could you give me a more time to digest this fragment?** And then after that, I just choose the three of the button, my race full board, revivals, and then when I click the button, and they also have a really different and little bit lot of information for me and so Let me think so this application is for children, not for others.

Speaker 2 (06:16): And so, yeah, so I guess that **this application is really familiar with adult. Because we have a lot of experience using the game,** especially in the fighting summon. Like I competition with someone, **but as a child as a children's, maybe they are not familiar with this interface.**

Speaker 2 (06:42): I mean, so yeah, so I just worry about that.

Speaker 1 (06:48): Uh go to the shop and Explore it.

Speaker 2 (06:55): Okay. Ah, society right now I have strength. So these 3 coins and the 11 crystal, but I'm in poor I don't get anything, but it's really clear to get, I do not get anything because your buttons is disabled.

Speaker 2 (07:15): So this ux is really easy to oh I cannot get that. Because my coins, yeah, my resources to really feel less than they require the HR attempts. Okay, yeah, okay, yeah.

Speaker 1 (07:29): Go to the theme.

Speaker 2 (07:32): Theme.

Speaker 1 (07:33): Change the interface to dark mode okay. explore the leaderboard again. Explore the shop.

Speaker 2 (07:41): mmmm oh I see.

Speaker 2 (07:45): I see. I think okay.

Speaker 1 (07:53): Just play again

Speaker 2 (07:57): Play again. Okay.

Speaker 2 (08:09): Flag. A flag. Oh, flag.

Speaker 2 (08:18): Hey Flag. Flag. Flag.

Speaker 2 (08:30): Hey, okay.

Speaker 1 (08:38): Okay, um, again, go back to the theme switch the Game mode.

Speaker 2 (08:44): Day?, oh yes. ah okay.

Speaker 1 (08:51): And then, after switching, go back to the game interface and play again. Just pick any

Speaker 2 (08:59): Okay. No, maybe I guess this time still love castle. So I need to choose active button.

Speaker 2 (09:06): Okay? Hmm, yeah, got it. Stop.

Speaker 2 (09:15): Yeah, these times I just choose to stop mode. That's okay. Hi hahaha Hey But it works.

Speaker 2 (09:34): Stop, stop, stop. Hmm, interesting in okay. I'm wonderful.

Speaker 2 (09:45): It's

Speaker 1 (09:45): Alright. So if you're comfortable with any of those theme, you can either go back to light or dark, whichever you're comfortable with.

Speaker 2 (09:55): Yeah, I like the dark mode, alright.

Speaker 1 (09:58): Go to settings.

Speaker 2 (10:00): Setting, where's the setting home? Okay?

Speaker 1 (10:04): I go to settings and then change the orientation. Yeah, the layout orientation to landscape. Oh, play a game in this layout.

Speaker 2 (10:21): I don't need to go back Yeah, yeah.

Speaker 1 (10:34): Okay just scroll and play a game

Speaker 2 (10:43): This time I just choose to Easy Mode in the clap. Oh, yeah. Oh, yeah, right.

Speaker 2 (10:57): Hey, Clap. Clap. Clap.

Speaker 2 (11:08): Did you hear me? Clap. Maybe this is the problem of my pronunciation.

Speaker 2 (11:19): Okay.

Speaker 1 (11:23): So um play another game.

Speaker 2 (11:26): Okay so this time, I'm gonna dragon in hard no Easy Mode. Oh, the dragon. Dragon.

Speaker 2 (11:40): Dragon. Wow, dragon. Dragon.

Speaker 2 (11:57): It's really Ok, dragon. Oh, okay? So okay.

Speaker 2 (12:09): Can I give you some kind of opinion in this landscape? Yeah, so I originally used my right hand and then like portrait mode. That time the how can I say the controlling buttons?

Speaker 2 (12:28): I mean, the top middle and the bottom. That thing is the right side. So it's really a handful for to controlling that avatar.

Speaker 2 (12:38): but in the landscape mode, the controlling buttons they are on the left side. And then it's a little bit hard to controlling that, but I know this. Maybe the monsters there from the right side of you put that buttons on the left one.

Speaker 2 (12:57): But yeah, it's a little bit hard to manipulate in my outcome.

Speaker 1 (13:03): Okay go to the setting.

Speaker 2 (13:05): Setting.

Speaker 1 (13:06): and, you can just shift to potrait,. Go to the settings, increase the text size.

Speaker 2 (13:16): Okay? And increase the text size, oh yeah.

Speaker 1 (13:18): It was already increased? Okay, then go to the world display and change the word display.

Speaker 2 (13:28): The word display.

Speaker 1 (13:31): you can increase the text size to a point where you can see

Speaker 2 (13:37): So at the first my setting is really biggest one. So this time, I changed the small the most small one okay. Biggest one okay.

Speaker 1 (13:50): And then go to the word display, pick any change it

Speaker 2 (13:53): okay so this time, I just want to choose on enemies.

Speaker 1 (14:01): And then change the game duration. Also

Speaker 2 (14:08): And also, I'm gonna pick up 1 minute, okay.

Speaker 1 (14:11): And then play a game

Speaker 2 (14:21): Ah, frog.

Speaker 2 (14:22): Frog. A frog frog frog frog. Frog.

Speaker 2 (14:34): Frog. Frog. Frog.

Speaker 2 (14:44): The frog. Frog. A frog.

Speaker 2 (14:54): Ok, and so this time The word, what I have to speak. It is on their head. I mean, on the monster.

Speaker 2 (15:07): It's really easy to catch. what I have to speech and then also I don't know why. But it's more feeling me feel more comfortable to catch what I do, so I guess if I were a developer I do.

Speaker 2 (15:26): this mode, this second mode on the enemy is on the default mode. So yeah.

Speaker 1 (15:33): Wonderful, okay, yeah. So, go back to settings. And then in the settings, go to contrast and turn it on

Speaker 2 (15:45): That's okay. So yeah.

Speaker 1 (15:47): And then go back to the game and play

Speaker 2 (15:57): Okay, clap, clap, cla. Clap. Clap.

Speaker 2 (16:07): Clap.

Speaker 2 (16:12): I like this mode so right now I play and I'm playing now. In contrast mode, and then in this mode, I finally, cut it to the left side in the game. What I mean before this mode.

Speaker 2 (16:30): I do not know if the high mid low I mean, I just focused on the monsters and the coins, but in the contrast mode I can catch the where I am. Because I can see the left left side is interruption. Yeah, so it is really clear to Oh, the contrast means.

Speaker 2 (16:58): Yeah like this. Like it.

Speaker 1 (17:04): Okay, so that's it for the um child interface, and so what we're going to do. We're going to move on to the dashboard

Speaker 2 (17:13): okay.

Speaker 1 (17:15): So the password is 0000?

Speaker 2 (17:18): Okay. Okay.

Speaker 1 (17:20): Okay? So if you look at the accuracy for the game, the session you just played, okay. Just navigate that tab, just look at the progress tab first.

Speaker 1 (17:33): We're just gonna start with the progress tab, so just yeah, so just explore that tab look at the accuracy of all the session. that you just play so just scroll and see it, and then what I need you to do is to Go to the progress report. Yes, the progress report, and just show me how you share it.

Speaker 2 (18:00): Ads on there So I take care. Take care like a parent. Yeah so like

Speaker 1 (18:08): As a parent, So just go to that progress report

Speaker 2 (18:15): Okay, AM. Yeah, it was really nice. Because it is really easy to catch.

Speaker 2 (18:22): Which word is really good? So for my kids or which word is really less. They have more practice.

Speaker 2 (18:32): And also, the one thing is really attractive for me. Is you gave PDF button? Because let me think if I were the caretaker.

Speaker 2 (18:45): If then maybe I need to share this information for like a doctor's or someone has really had medical Medical thing so maybe that for that is really clear easy to share my kid situation, state. So yeah, maybe if I were their parents. Maybe I will use a lot.

Speaker 2 (19:13): Maybe use this functionality pdf and also I'd like to record my kids process. So it's really clear, okay? Alright, um.

Speaker 1 (19:30): Ok, now go to the lesson tab, and I want you to scroll down and create a lesson.

Speaker 2 (19:39): Ok, ooh. Hmm. Name of lesson oh my Hmm, or like hello?

Speaker 3 (20:03): Bye.

Speaker 2 (20:04): Yes finish

Speaker 1 (20:06): and then I want you to activate that lesson. And then go to the game interface. And just look at the play.

Speaker 2 (20:22): Wow, it's really impressive, okay.

Speaker 1 (20:27): And then I want you to go back to the dashboard. And edit the lesson and then go back to the game interface and see

Speaker 2 (20:40): Oh

Speaker 1 (21:03): And then you can delete the lesson

Speaker 2 (21:15): Okay.

Speaker 1 (21:16): Okay, so still in the lesson. Change the practice mode.

Speaker 2 (21:24): Change the practice mode, and how can i get there. Hmm

Speaker 1 (21:32): Change the game mode sorry, change the Game mode.

Speaker 2 (21:38): Okay, I picked ah, choose your word mode then.

Speaker 1 (21:44): Then go back to the interface and just see

Speaker 2 (21:52): change the game mode. just to see

Speaker 2 (22:17): Hey, hey, okay. All right. got it

Speaker 1 (22:22): And then I need you to reset the password for the clinical interface.

Speaker 2 (22:27): Password, where I can go.

Speaker 1 (22:31): Just go back

Speaker 2 (22:32): Settings? dashboard, ah, okay?

Speaker 1 (22:41): Yes, it's okay, just click

Speaker 2 (22:47): Ok.

Speaker 2 (22:56): Ah, yeah, ah.

Speaker 1 (23:01): okay just try your new password?

Speaker 2 (23:07): Yeah, it works.

Speaker 1 (23:09): All right. Um, now I just need you to explore the rest of the application in just experiment with it. So just test everything that you fully haven't test.

Speaker 1 (23:17): Okay? Hey, yeah.

Speaker 2 (23:20): Haptic feedback

Speaker 1 (23:33): You can try the haptic if its on

Speaker 2 (23:35): Ok. Clap clap dragon. stop.

Speaker 2 (23:53): Ah, okay, clap. Clap. Got it, okay.

Speaker 2 (24:12): Okay.

Speaker 1 (24:13): just explore everything.

Speaker 2 (24:17): Finally, I can get iron shield. Okay, Lets, just try this shield. How can I use this?

Speaker 2 (24:30): Is this oh flag? Dragon. frog.

Speaker 2 (25:01): Okay.

Speaker 1 (25:05): You feel comfortable have you explored everything even in the dashboard? Alright.

Speaker 2 (25:21): Okay.

Speaker 2 (25:24): I'm gonna let the first of all, this application is really impressive. Because there are a lot of functionality in here.

Speaker 1 (25:34): Okay we are done with the think aloud. So what we're gonna do, we're going to move on to the semi structured interview. And then from there get your opinion

Speaker 2 (25:46): We'll get your opinion okay.

Speaker 1 (25:50): So yeah, we're moving on to the structured interview. The semi structured. Are you ready?

Speaker 2 (25:54): Yep

Speaker 1 (25:55): Are you ready? Yeah, all right, so Alright. So first question here Um, during the gameplay or during the tasks I noticed you would stop pronouncing the words and then you would lean forward to pronounce.

Speaker 1 (26:13): It was there a reason for it. you'd pronounce it, and then you'll lean forward and pronounce louder to the microphone.

Speaker 2 (26:24): Honestly, when I used the this game, I just focused on how to move the monsters and just follow the coins. So I mean, it's a little bit less pay attention to the how can I turn out this word? Or this application is really developed to catch my words.

Speaker 2 (26:44): So it's a little bit out of mind, thinking, yeah.

Speaker 1 (26:49): All right, I also noticed you preferred the dark mode,

Speaker 2 (26:52): Yeah, exactly.

Speaker 1 (26:54): was there a reason for it or just your preference

Speaker 2 (26:57): Because I maybe I guess it's just for my personal taste. But also for me, the Light mode I mean day mode is they have too many colors, so it is really hard to catch what they're in it. So I mean, in the Dark mode, it's more made me feel comfortable to catch their information.

Speaker 2 (27:22): And then using this game. So yeah, it shows them like light mode for me, looks too many information, and at 1 time. So that's why I use that

Speaker 1 (27:37): All right, so how would you describe SPOLVE which is the application to a parent of a child who has cast classes like a speech disability? So how would you describe SPOLVE to them? and they've never seen a speech therapy app before, how would you describe this up?

Speaker 1 (27:59): And what would be your concerns for it to me? Start with how you describe it first. or what you to think the app is

Speaker 2 (28:11): As in their shoes,

Speaker 1 (28:15): no, no, just you. So Let's say. I'm the parent and you've already interacted with the app so I want you to describe the app to me.

Speaker 2 (28:28): I would describe this

Speaker 2 (28:36): Yeah, this app is developed for the kids who have some struggle with speaking. But I guess lots of application for rehabilitation is just focused on speech. I mean, a lot of application push kids and maybe it's going to be a mixed kids really burdensome.

Speaker 2 (29:01): So maybe it's going to be they make they do not want to use that kind of application because they do not feel any interesting or joyful about that. But this application is a little bit different, because as I said, in the first question, when you talk about when you ask about speech thing and the time I answered, oh I do not care because I just enjoyed the game. So that's what I can tell.

Speaker 2 (29:30): This application is really good to encourage the kids. Using this application, ok, okay?

Speaker 1 (29:38): And what would be your main concerns about the application?

Speaker 2 (29:41): main concern. I can't tell the main question is Kids. May they have too much upsets with the game?

Speaker 2 (30:00): I mean, in the game, they always have a control button. I mean, they can move their Avatar without their speeching. They do nothing, maybe they can.

Speaker 2 (30:12): They will be able to do nothing about like. Okay, not no more speech and just controlling the avatar to avoid them. So, yeah, they still want them concerned about that.

Speaker 1 (30:25): Wonderful. Okay, do you think any gamification elements? For example, the leaderboard could be counterproductive.

Speaker 1 (30:34): For a child

Speaker 2 (30:34): Ah

Speaker 1 (30:36): Already anxious or self conscious about their speech

Speaker 2 (30:44): Yeah good question, because and I honestly, it's a really personal opinion. I do not have any Interesting competition. So if I were a kid, I do not care about.

Speaker 2 (31:01): Oh, my rivals. Also, who cares? I'm not care, but But most of all kids, they really think about the others.

Speaker 2 (31:13): Their peers, so maybe they have I know, so maybe they have Some kind of stress from them, because I For me, rivals means like I have to fight, and I had to take down them. So it makes me feel like a little bit negative stress. Yeah, so yeah.

Speaker 2 (31:46): So I mean maybe full board, which is which I can catch the I can list I can see the list who is the other users in this app is enough and I mean rival is a little bit too far, yeah.

Speaker 1 (32:05): Um ok. So if you could only keep 3 features

Speaker 2 (32:11): Okay, 3 features.

Speaker 1 (32:12): Not to remove everything else. Which 3 would you keep? And why

Speaker 2 (32:19): As a parent's or kids

Speaker 1 (32:22): So you'd start with kids. Okay

Speaker 2 (32:25): So kids. Um, if I were kids. I would like to keep Of course.

Speaker 2 (32:34): One is Absolutely Game mode and also in the Hmm, controlling buttons and also . Okay, there is the item shop and then last one is

Speaker 2 (33:02): Last one is that one difficulty level, yeah?

Speaker 1 (33:10): And a on the speech therapy board on the dashboard, three things you would keep.

Speaker 2 (33:18): Dashboard. Yeah, of course. For the first one is definitely Pdf button.

Speaker 2 (33:30): This is the thing I guess, and the second one is This calendar because it's really easy to catch day their achievement because I think if I were parents. It's really yes, keep the streak. It's really important so that then means maybe parents really want to check their streak

Speaker 2 (33:56): And then this, yeah, this design, I mean, the calendar is really easy to catch their achievement streak, and then last one is Lessons, I mean Create a lesson. This one is really important because, I guess each kid they have their own struggling words. So give them personalizing mode is really important.

Speaker 2 (34:29): I guess so I choose that.

Speaker 1 (34:32): I was good, alright. Is that anything about SPOLVE? That surprised you positively or negatively.

Speaker 1 (34:40): That you have not yet mentioned

Speaker 2 (34:48): Oh, yeah. so yeah. The opinion which I didn't mention yet is yeah, I have it, but I am a little bit.

Speaker 2 (35:01): I forgot it so could you give me a second.

Speaker 2 (35:11): In the ux way, I'm a little bit. I don't like waste time and then that's mean I really moved fast. So maybe that's why I feel like some kind of some touch.

Speaker 2 (35:23): Some click is feel like a little bit. I'm too elaborate. I mean, I just try to click it and not click it just like I'm trying to hovering it.

Speaker 2 (35:38): But it works so it made me feel. Oh, this application is more than faster. I guess I expected.

Speaker 2 (35:46): So. Yeah, you UX view. And then next one is hmm.

Speaker 2 (35:57): Yeah, I can feel the developer. You are willing to think of this application by users' view. Because you already have a real impressive functionality.

Speaker 2 (36:12): In the Game mode, even in theme and then also yeah, in the setting mode. , you gave the user lot to choice, and it was really impressive. Then, but if I were a kid, as I said in the beginning of the interview I mentioned, if I were a kid, I'm a bit confused because there are lots of functionality in the one application. I mean there's one screen, I can catch lots of information.

Speaker 2 (36:48): So this is really good for Tamm dam to allow them free. And not on the other side, he's maybe taken confused because there are lots of things. So maybe you'd better, how about this you just are just like compact mode.

Speaker 2 (37:10): Maybe there's some kind of kids needs that, because maybe kids they just want to play in the game. Yeah, anyway, yeah, this was really impressive. I don't know, so there gives us a lot of free choices from the avatars.

Speaker 2 (37:28): Even yeah, yeah, avatars, you can give me a lot of choices to choose skins and then profile images. Yeah, from the there was yeah, that was the first impression impression when I used this application and that was really nice. Oh, yeah, so one more thing, sorry, this application is designed.

Speaker 2 (37:53): I mean, color system, it's really cozy. Okay, so I can. And even in both mode, a Dark mode and Light mode, even though I love the Dark mode.

Speaker 2 (38:04): And so it's real, I can feel, oh, it's really fun really familiar with kids. Hso I can find developer or designer. They really take care of kids' view.

Speaker 2 (38:17): So this is really warm. I like that.

Speaker 1 (38:20): Thats lovely, do you think the application could be used with other people other than just kids? To be used with someone older

Speaker 2 (38:34): So your question is, maybe the other people

Speaker 1 (38:37): Yeah, can it be used by other people other than kids? Maybe they see the people have other speech impairments. So, could it be used by other people other than kids?

Speaker 2 (38:52): Yeah, exactly they really loving this. Yeah, I think this application is in the production mode.

Speaker 1 (39:00): Oh, God that was great. That's it for the questions you're done. Great job.

Speaker 2 (39:07): Yes, I really like it, yeah.

APPENDIX B- Interview Transcript 2

Participant 2 = Speaker 1

Speaker 1 (00:00): 18 to 3 within the boundaries. For this. So see, alright, okay?

Speaker 2 (00:07): And then okay?

Speaker 1 (00:08): So I'll opt the code character. I guess the only thing is, as I put my thumb down. It sort of picking one I want to select that and keep it as I put my thumb down, it sort of scrolls the I gotta put my thumb between these to select it.

Speaker 1 (00:25): I want to okay I see, there's an equip button. So that works. Okay?

Speaker 1 (00:34): I want to proceed. Are there okay? There's a save Avatar in the edit Avatar thing.

Speaker 1 (00:40): So I think if I go back to now, I'll be okay So I'll start playing, okay.

Speaker 2 (00:48): And feel free to adjust the difficulty modes.

Speaker 1 (00:51): Okay, I'll do. I'll start with it easy for now. Okay, flag.

Speaker 1 (00:59): So I was a flag Oh, I see. So it's like mid flag. Okay, low so flag.

Speaker 1 (01:15): Have I got to? There's buttons to move on the side here flag. I can't let low Oh, okay, they said it looked like there was going to be an enemy, but then it passed me and it was free stars, but they said a 100 actually, before I missed 1 or 2.

Speaker 1 (01:38): So if I go to another word, stop. Oh, that's okay. I feel like it was mid stop.

Speaker 1 (01:51): Stop. Okay, hi, an enemy just disappeared stop. Power blast stop.

Speaker 1 (02:02): I'm not particularly sure what a power blast is doing. Stop Stop. Okay, I think Okay, I'm curious about what coins and stars do.

Speaker 2 (02:19): let me know when you're done with the gameplay and then we'll move on to the next bit for it

Speaker 1 (02:23): Yeah, I think I'll try. I try it hard for one

Speaker 1 (02:30): Uh, okay, frog frog. Frog. Yeah, frog?

Speaker 1 (02:40): Frog. Frog. Frog.

Speaker 1 (02:45): A frog. A frog. Frog.

Speaker 1 (02:53): Frog. Frog. Alright, frog.

Speaker 1 (03:02): Yeah, I felt like there. It wasn't processing my speech fast enough compared to how many enemies were appearing on screen at once. I felt like I wasn't even I was pronouncing the word it didn't feel like I was getting much feedback out of it.

Speaker 1 (03:17): So. I've tried the three medium free difficult. The three difficulties three of the four words. Should I do one more word or

Speaker 2 (03:27): uhm yes.

Speaker 1 (03:34): I'll try Clap. Clap. Clap.

Speaker 1 (03:40): Clap. Clap. Clap.

Speaker 1 (03:49): A clap. I always admit clapp. Clap, clap.

Speaker 1 (04:02): Clap. I lost a life. There was actually okay, yeah, that's all the words done.

Speaker 2 (04:09): Okay, go to the leaderboard and then see the rankings.

Speaker 1 (04:15): Okay. I am 30th.

Speaker 2 (04:18): Okay. Just explore that

Speaker 1 (04:22): So Your race, I'm not sure what race dictates a full board. Okay, there's a lot of users. rivals, I didn't know I had competition.

Speaker 1 (04:39): Okay, people played now interesting. So, kind of click on that See is full profiles. No, it's just It's interesting.

Speaker 1 (04:55): Um, uh, U. Okay, this looks like a scrolling banner, but I can't control the speed of the messages it just goes on quite a while. Oh, there's a key at the bottom.

Speaker 1 (05:09): Okay, so playing now on the street rival within 200 points of your score interesting, so okay? Is there more to do on the leaderboard or?

Speaker 2 (05:26): Um no, so you can move on to the shop.

Speaker 1 (05:29): Okay, I'm going to the shop. Okay? This is where I spend my coins.

Speaker 1 (05:33): I see, there's also crystals, so I think I'll, I don't have enough coins for many of these. I think I'll keep scrolling just to see How can iron plate And that's what I can afford currently. So I'll go back And so I start playing with the new item, I got So I'll try So I made a flag okay flag.

Speaker 1 (06:13): Flag. A flag. Flag.

Speaker 1 (06:29): Flag. It's like So it seems like I didn't need to use the item there actually low difficulty. So I guess I'll go through medium

Speaker 1 (06:50): flag. maybe its best if I do hard difficulty for then sort of incite the item to be used faster just because there's more enemy spawning at once on the high difficulties so I'll do that.

Speaker 1 (07:08): Let's see flag. Oh, it's just slower than before. The flag Flag, okay, well.

Speaker 1 (07:24): It's a week. Flag. so that's strange.

Speaker 1 (07:34): It's a massive slowdown on the app. As well

Speaker 2 (07:36): Yeah

Speaker 1 (07:39): Okay, okay, okay. Uh As soon as I did, I'll go back to the shop and see what so ads + 1 heart, I did notice that did give me an extra heart. But I had the armor, the stopped me from losing a heart.

Speaker 1 (07:53): So I didn't get to see that in action. But yeah, the high difficulty also then wasn't focusing as well as it was before. At least so that it was a similar speed, easy medium.

Speaker 1 (08:05): But then there were less enemies. I was confused what was happening because I'd had used the hard difficulty before so I'll go back to the menu.

Speaker 2 (08:13): Okay, um hm. Alright, so go to the theme and switch the interface to Dark mode. And then, okay, trying to play from there

Speaker 1 (08:28): I'll play again, do

Speaker 2 (08:29): Yeah, it just explore it from the dark mode

Speaker 1 (08:33): Okay. Stop. Stop. Stop.

Speaker 1 (08:42): stop. stop. Stop.

Speaker 1 (08:54): Stop. Stop.

Speaker 2 (09:05): Yeah, just from the dark mode, explore the other places like the shop and the leaderboard just to see

Speaker 1 (09:13): I see that a has consistently overhauled all of the menus to be night mode. yeah It seems every Every menu has been changed its good.

Speaker 1 (09:32): Uh.

Speaker 2 (09:36): You can switch the theme

Speaker 1 (09:38): Okay, yeah, I'll switch theme. I've seen enchanted forest that was one. I've missed, so I'll so then I'll play a new game.

Speaker 1 (09:46): I'll do This one.

Speaker 1 (09:51): ah, okay? Frog, a frog frog. Frog.

Speaker 1 (09:57): Frog. A frog. Frog.

Speaker 1 (10:04): Frog. Frog. Frog.

Speaker 1 (10:10): Frog. A frog. A frog.

Speaker 1 (10:17): The frog. frog.

Speaker 1 (10:24): I noticed that the shield has expired after 2 games

Speaker 2 (10:33): Wonderful, alright, so I need you to go to the settings.

Speaker 1 (10:39): Away from the screen onto the homepage

Speaker 2 (10:42): Yes

Speaker 1 (10:46): I see, alright. It's just I just tell you. So I see settings for game duration.

Speaker 1 (10:55): So this goes on 30 second increments. Yeah, there's accuracy captions in game hints. I'm not sure where accuracy captions were being displayed.

Speaker 1 (11:07): Just because the game was running quite fast in game hints and also unsure what that hints those were the word selected instructions. I think that maybe needs to be more clarifications as well. Word display.

Speaker 1 (11:24): I think it's fair and they did show it did display a word above the target. So I mean I knew what word to say Oh, I can. Okay?

Speaker 1 (11:32): So if I type change words by okay, this is a toggle. So I can say a banner on top of the enemies, which is how it was before it's default or above the microphone only I switch this to banner.

Speaker 2 (11:47): okay.

Speaker 1 (11:49): So I can change the inside the controls haptic feedback. I have not felt haptic feedback so far.

Speaker 2 (11:55): okay

Speaker 1 (11:57): A high contrast mode, turn it on. Oh, okay, I've changed the text, and it affects the text. But it also warps the menu when I do it.

Speaker 1 (12:08): So it is a bit disorientating.

Speaker 2 (12:17): So I want you to change some of those settings as well as the orientation, and then, yeah, feel free to play

Speaker 1 (12:26): Yes, yes, I see. Auto rotate the landscape. I will try rotate the landscape

Speaker 2 (12:33): So before that, we use set it to haptic, can you just put on top and see if I put that on vibration? I'm sorry, I just have to take I'm sorry, yeah, thank you. Oh, I'm sorry.

Speaker 1 (12:50): Okay, okay, this time I'll set it to landscape.

Speaker 2 (12:55): I also feel free if you prefer Light mode or Dark mode.

Speaker 1 (12:59): I don't mind so I noticed although a lot of the elements are displayed in landscape. The character, the play character, sort of cropped out doesn't fully show Even If I scroll up. Yeah, I think sorry I'll click play now.

Speaker 1 (13:19): I feel like If I launch the game in landscape, I would only really be clued in that there were options below. If I scroll down. But because I'd seen in vertical, I know that they were here so quick stop by I feel like This is also very zoomed in, so if I do flag Play okay, flag.

Speaker 1 (13:56): flag Flag.flag flag flag flag flag flag. Oh, okay.

Speaker 1 (14:23): So when I changed the landscape around, I've noticed that I can see more of what's going on at once it gives me more time to process what's coming towards me. So which then gives me more time for the words to actually process in the game.

Speaker 1 (14:41): I'm curious about if the game will adjust accordingly. If I swap rotation and say accidentally mid game so for the flag. Alright, so okay.

Speaker 1 (14:53): So there's more going on outside of flag. There's more going on outside, there's just not in view, interesting It's like I feel like I have an easier experience when I'm playing It in landscape like It's like I can now rank 23. I've been glad tells me I've been climbing the board.

Speaker 1 (15:26): Okay. so what else do you want me to do.

Speaker 2 (15:34): Okay, so Did you activate the High contrast?

Speaker 1 (15:39): Oh, yes, I think I did activate High contrast yeah?

Speaker 2 (15:44): Okay, so Now we're going to move to the next part, which is the clinical dashboard. So I just need you to go back and then go to the dashboard.

Speaker 1 (15:58): dashboard.

Speaker 2 (15:59): And then sorry, password is 0000 if that doesn't work just reset it And then I'm just look at the accuracy data from the session. You just played

Speaker 1 (16:13): 92% is practicable practice time spent total attempts Okay, so there's a way different from today. Which? Okay.

Speaker 1 (16:34): Choose your word full lesson mode. Okay, so I have a choice on which mode. Uh, I will.

Speaker 1 (16:45): change the mode

Speaker 2 (16:47): So before we just go to that just in the progress tab, Can you find the progress report?

Speaker 1 (16:58): The progress report

Speaker 2 (17:01): Find it and then just click to see how you'd share it.

Speaker 1 (17:05): Uh, downloads is a PDF, and then it opens the PDF record.

Speaker 2 (17:10): That's perfect, and then I need you to go now to the lesson board or less on tab. So create a lesson

Speaker 1 (17:24): new lesson

Speaker 2 (17:25): Then, after you've created, just activate it

Speaker 1 (17:28): Okay, so three words, so this is the name of the lesson I'll just do new lesson new lesson, okay. Next pick words. It's a type of search out of the word.

Speaker 1 (17:45): Let's see, let's do a dog. Okay, so it says it doesn't exist. So add it, let's see Let's just do an animal themed one.

Speaker 1 (18:02): Let's just do cat then. I'm trying to think all what other words could be so we've got cat.

Speaker 1 (18:18): Oh, keyboard lets do lizard. that's three words there. I think I accidentally tapped finish

Speaker 1 (18:28): But I think the word is there. Yes, new lesson. Okay.

Speaker 1 (18:30): So if I set this to active, yeah?

Speaker 2 (18:34): yes, then I'll go back to the game interface. Just to see

Speaker 1 (18:42): All three words, okay. So this is gonna mix between them. And this is, these are the words that I added, okay.

Speaker 1 (18:50): A dog dog. Lizard. Lizard.

Speaker 1 (19:06): Oh, God, sorry, cat

Speaker 2 (19:11): Okay. I was completely fine.

Speaker 1 (19:18): The interest thing I feel like it's not doing. It's not recognizing single syllable words. Yes.

Speaker 2 (19:26): Yes, I'm glad you picked up on that. Okay, because you already finished the child's interview. So it's completely fine.

Speaker 2 (19:34): So you just go back to the dashboard. Then, feel free to edit the words in your lesson. As well as delete it, okay.

Speaker 1 (19:49): So the words So lessons It's like so I can edit it okay, so

Speaker 2 (19:59): Good so when you edit just go and confirm it in the game interface, you don't need to play It, just confirm then you can come back and continue

Speaker 1 (20:08): and so then I can also add words that were already defined. Does this mean then that I could create another lesson? And import those words from the previous lesson that I'd created

Speaker 2 (20:22): You could try.

Speaker 1 (20:23): Let's see, there's 5 Ah, no it does recognize the words from the last lesson. It still says cat is a custom word. Okay.

Speaker 1 (20:38): I'll go back to the old lesson and delete a word? And then now go back to the game, just to see if it's Actually, the easiest thing to do would probably be to change the gameplay to, which I was going to, yeah choose your words that I could Quite clearly then see if that has deleted the word. Yes, it has okay.

Speaker 1 (21:10): Okay.

Speaker 2 (21:13): Okay. Can you reset the password?

Speaker 1 (21:17): Reset? Oh yes, okay. So I said forgot pin, oh no, that's not what I want.

Speaker 2 (21:25): So yeah.

Speaker 1 (21:27): I did no This is the second thing. Also, the pin is here if I was to send an email

Speaker 1 (21:49): Okay reset

Speaker 1 (21:59): 1234 okay.let me double check Yeah, there we go.

Speaker 2 (22:03): then I just want you to explore the rest of the application feel free to play around with it.

Speaker 1 (22:07): Okay. Also, I've seen the session calendar. The words so I can So I noticed that there's a disconnect between words and lessons, there's words in the lessons, but then there's these active words.

Speaker 1 (22:25): But if I have a lesson active, would those be the active words? If I search for Yeah, the words from the lessons I've created. And aren't appearing here in the words section, either.

Speaker 1 (22:45): Let's see. If I delete. So I'm gonna go to the lesson I created

Speaker 1 (22:57): I delete it. I'm curious if it'll have an active lesson if I've deleted the lesson there was previously active so I'm going to go back and check and yes, a default to the active lesson. That's a that's good.

Speaker 1 (23:21): Okay, um, I'm curious about combined mode. So I'll go and You see that? I also noticed.

Speaker 1 (23:32): I can edit the profile, so I can change the age, but even though when creating the profile, I had an option to change the profile picture. There's no option for me to do that again. This seems permanently associated with the profile like You see, if we start, oh Okay, so it seems that this gives me option to both.

Speaker 1 (23:56): Do all four words, or just pick a word. So let's see, clap. Flag.

Speaker 1 (24:10): Flag. Frog. Clap Nope, uh, frog.

Speaker 1 (24:29): Stop. I've noticed that my time extended there. But then it ended And there were more enemies on screen, and I thought I would be playing for longer, but yeah, too close.

Speaker 1 (24:47): Okay, was there more you want me to do or

Speaker 2 (24:51): Um. No, you actually did really good. That was, was it?

Speaker 2 (24:57): I just wanted to see if you wanted to see any more things apart from what I asked you to do that, you just want to try for yourself.

Speaker 1 (25:04): I think I was quite thorough there. I think

Speaker 2 (25:06): Okay, that was wonderful. Then, that's the end of the think aloud. So we just kind of move on to the interview questions.

Speaker 2 (25:17): Okay so when you are creating the lessons I noticed, you mentioned being able to import from a previous lesson, import words from a previous lesson. Can you tell me more about what you were thinking?

Speaker 1 (25:31): I noticed that there was a search functionality for the words, and I was under the impression that that would let you then pick words that were established in previous lessons. If because I'd put in dog and cat and have noticed the game, then noticed that those weren't pre-existing words like create those. So I thought that they would be able to search back and find those words

Speaker 2 (25:55): Okay. I'm wonderful.

Speaker 2 (25:58): Let's see if you search like the. Built-in words, were you able to see them?

Speaker 1 (26:05): Yes, yes, I want yes.

Speaker 2 (26:06): Alright. So how would you describe spolve after playing it to a parent of a child who's never seen a speech therapy up before? Like what would be your main concerns, or how would you for example, describe it?

Speaker 1 (26:23): So I would describe it as I came the incentivizes, sort of learning the words sort of saying I'm in a time in manner. There's a lot. There's a big focus on sort of like a time limit where the enemies are approaching you and you have to say the word in time.

Speaker 1 (26:44): And they are also very short play sessions. So that promotes quite a lot of experimentation with the words because there are multiple different words to choose from and you can create new word sets and then mix a match with if you want to just practice one word in a short burst of multiple.

Speaker 2 (27:03): Okay, and what will be your main concerns?

Speaker 1 (27:06): So I I'm unsure, so I don't have any speech issues, but if given this to a child, I'd be concerned that this wouldn't be giving them sort of urgency or like for them to take their time with pronouncing these words. It's sort of especially with the harder difficulty, why it sort of if at the point where they could play on the hard difficulty consistently. It sort of I'm much rather worried, but it's like it's they should.

Speaker 1 (27:46): Is there a point for a hard mode where like they can say the words that faster at that point? For a sort of I have a losses. I suppose um.

Speaker 1 (28:00): It's sort of also. There's just the issue of with the speech process, so it's sort of If it felt like I was saying the words in time, but then there was some sort of gap where it would process it. But then it would go to the next enemy and then it would then finish processing the word and get the enemy behind.

Speaker 1 (28:25): So I wonder if it's sort of wouldn't give accuracy feedback to the trial and the pronunciation sort of I wonder if that could perhaps be an issue where a child, perhaps mispronounces word, it still processing, they then mispronounce the word. But then, like the feedback gives them something positive. Because there's another enemy that then appears, and then is killed.

Speaker 1 (28:57): So

Speaker 2 (29:02): So if maybe during its actual development, we used a better model for the speech recognition would that cater to some of your concerns.

Speaker 1 (29:18): But yes, if the model was able to handle fix the issues of responsiveness and was also more lenient with a single syllable words I felt like I tried to pronounce words quite clearly, but it wasn't. Recognizing me that well, yeah. So if the model was able to address those issues, that would be a big benefit, I think

Speaker 2 (29:47): Wonderful. Okay, so do you think any of the gamification elements? For example, the leaderboard could be counterproductive for a child who is already anxious or self-conscious about their speech.

Speaker 1 (29:59): I feel like a global leaderboard with sort of a difference in scores like that drastic would perhaps be quite intimidating for a child. At first I think a local leaderboard or sort of a kid in clinic or what sort of administered at the same time to therapy would probably be better to sort of if you have kids to only a couple weeks apart and their progress compared to sort of a child opening app for the first time, seeing that there's someone with 200000 points. That would be quite intimidating.

Speaker 1 (30:37): I think

Speaker 2 (30:39): It's actually very good, that's a good point. If you could only keep 3 features of spolve and you had to remove everything else. Which 3 would you keep and

Speaker 2 (30:52): Why?

Speaker 1 (30:54): I like the dashboard, the dashboard with the custom words. I think that that'd be quite good for sort of custom, tailoring therapy to a child after having issues with certain words, it's good that they also addresses which words would be of concern and generates a report so that'll be quite good. I suppose also I like the I suppose the presentation the man used.

Speaker 1 (31:25): I like how clear the buttons are. It's a sort of I know, I guess I like the controls the idea of like sort of moving up-and-down, it sort of creates A. It's not just sitting there and saying the word is sort of it's more of a gameplay element to it where you're moving up-and-down.

Speaker 1 (31:48): And there's more to manage with that sort of Yeah

Speaker 2 (31:57): Okay. That was great. So is there anything about spolve that surprised you positively or negatively?

Speaker 2 (32:06): That you have not yet mentioned

Speaker 1 (32:10): So I was just sort of surprised. If how difficult the Hard mode was it was quite enemies spawning, quite fast. And there were a lot of enemies at once I think that the speech process thing was a surprise as well.

Speaker 1 (32:29): Oh, what else is there. Yes, I was quite impressed with the being able to switch between portrait landscape mode. If you like that, that'd be quite helpful for the child, they may just accidentally sort of rotate the phone during gameplay.

Speaker 1 (32:43): But I did feel as if sort of landscape mode was a lot more beneficial to the gameplay than portrait mode just because there was a lot more going on on the screen than there was still happening in portrait mode but wasn't being displayed. So it felt like it was just being more honest and giving more information

Speaker 2 (33:07): Yeah I think that was actually good. And those are the only questions I had for you and getting in depth. Yeah, I think we're done with the interview a bit. So I'm just gonna give you the questionnaire for you to fill

APPENDIX C - Analyst 2 Codes

ID	Excerpt	Code	Source (timestamp)
T1-S2-01	"There are a lot of information in this one fragment... could you give me more time to digest this fragment?"	Information overload in leaderboard interface	Transcript 1, Speaker 2 (05:25)
T1-S2-02	"This application is really familiar with adult... but as a child... maybe	Adult-oriented game conventions may not suit children	Transcript 1, Speaker 2 (06:16)

	they are not familiar with this interface.”		
T1-S2-03	“It’s really clear... I do not get anything because your buttons is disabled.”	Disabled state clearly communicates unavailable purchase	Transcript 1, Speaker 2 (06:55–07:15)
T1-S2-04	“I like the dark mode... the Light mode... have too many colors, so it is really hard to catch what they’re in it.”	Dark mode reduces visual clutter	Transcript 1, Speaker 2 (26:57–27:22)
T1-S2-05	“In the landscape mode, the controlling buttons... are on the left side. And then it’s a little bit hard to controlling that.”	Landscape controls feel less comfortable due to placement	Transcript 1, Speaker 2 (12:09–12:57)
T1-S2-06	“The word... is on their head... on the monster. It’s really easy to catch what I have to speech.”	Word-on-enemy display supports attention and comprehension	Transcript 1, Speaker 2 (14:54–15:26)
T1-S2-07	“In contrast mode... I can catch where I am... it is really clear.”	High-contrast mode improves spatial awareness	Transcript 1, Speaker 2 (16:12–16:58)
T1-S2-08	“It is really easy to catch which word is really good... which word is really less. They have more practice.”	Progress report helps identify target words for practice	Transcript 1, Speaker 2 (18:15–18:32)
T1-S2-09	“You gave PDF button... easy to share my kid situation, state... maybe I will use a lot.”	PDF export supports parent/clinician communication	Transcript 1, Speaker 2 (18:32–19:13)
T1-S2-10	“I just focused on how to move the monsters and just follow the coins... less pay attention to... turn out this word.”	Gameplay can distract from pronunciation focus	Transcript 1, Speaker 2 (26:24–26:44)
T1-S2-11	“This app is developed for the kids who have some struggle with speaking... this application is really good to encourage the kids.”	Game-based therapy may reduce burden and encourage engagement	Transcript 1, Speaker 2 (28:36–29:30)
T1-S2-12	“Kids may... have too much upsets with the game... they can move their Avatar without their speaking.”	Concern that children may play without practising speech	Transcript 1, Speaker 2 (29:41–30:12)
T1-S2-13	“Rivals means... I have to fight... it makes me feel like a little bit negative stress.”	Competitive/rival framing may create negative pressure	Transcript 1, Speaker 2 (30:44–31:46)
T1-S2-14	“Game mode... controlling buttons... item shop... difficulty level.”	Core child-facing features are gameplay, control, rewards, and difficulty	Transcript 1, Speaker 2 (32:25–33:02)
T1-S2-15	“PDF button... calendar... Lessons... each kid they have their own struggling words... personalizing mode is really important.”	Dashboard value lies in reporting, streak tracking, and personalisation	Transcript 1, Speaker 2 (33:18–34:29)
T1-S2-16	“You gave the user lot to choice... but if I were a kid, I’m a bit confused... maybe... compact mode.”	Many options empower users but may overwhelm children	Transcript 1, Speaker 2 (35:57–37:10)

T1-S2-17	"The color system... it's really cozy... fun... familiar with kids... they really take care of kids' view."	Child-friendly visual design feels warm and considered	Transcript 1, Speaker 2 (37:28–38:17)
T2-S1-01	"As I put my thumb down... it sort of scrolls... I gotta put my thumb between these to select it."	Avatar selection causes touch/scroll friction	Transcript 2, Speaker 1 (00:08–00:25)
T2-S1-02	"I'm not particularly sure what a power blast is doing... I'm curious about what coins and stars do."	Game mechanics and rewards need clearer explanation	Transcript 2, Speaker 1 (01:51–02:02)
T2-S1-03	"It wasn't processing my speech fast enough compared to how many enemies were appearing on screen at once."	Speech recognition responsiveness limits gameplay flow	Transcript 2, Speaker 1 (03:02–03:17)
T2-S1-04	"Rivals, I didn't know I had competition... this looks like a scrolling banner, but I can't control the speed."	Leaderboard elements are unclear and partly uncontrollable	Transcript 2, Speaker 1 (04:22–05:09)
T2-S1-05	"I don't have enough coins... I made a flag... it seems like I didn't need to use the item."	Shop affordance understood but item usefulness remains uncertain	Transcript 2, Speaker 1 (05:29–06:29)
T2-S1-06	"It's a massive slowdown on the app... I was confused what was happening."	Performance slowdown creates confusion around difficulty/item behaviour	Transcript 2, Speaker 1 (07:24–08:05)
T2-S1-07	"Every menu has been changed... it's good."	Dark mode is consistently applied across menus	Transcript 2, Speaker 1 (09:13–09:32)
T2-S1-08	"Accuracy captions... in game hints... unsure what that hints... maybe needs more clarifications."	Settings terminology needs clearer meaning	Transcript 2, Speaker 1 (10:46–11:24)
T2-S1-09	"High contrast mode... affects the text. But it also warps the menu... a bit disorientating."	Accessibility setting improves contrast but disrupts layout	Transcript 2, Speaker 1 (11:57–12:08)
T2-S1-10	"If I launch the game in landscape... I can see more of what's going on at once... gives me more time to process."	Landscape mode improves visibility and processing time	Transcript 2, Speaker 1 (13:19–14:41)
T2-S1-11	"It felt like I have an easier experience when I'm playing it in landscape."	Landscape is experienced as easier than portrait	Transcript 2, Speaker 1 (14:53–15:26)
T2-S1-12	"Downloads is a PDF, and then it opens the PDF record."	Progress report export is discoverable	Transcript 2, Speaker 1 (16:58–17:10)
T2-S1-13	"Dog... it says it doesn't exist. So add it... animal themed one."	Custom word creation supports personalised lesson content	Transcript 2, Speaker 1 (17:28–18:28)
T2-S1-14	"It's not recognizing single syllable words."	Single-syllable words may be harder for speech recognition	Transcript 2, Speaker 1 (19:18–19:26)

T2-S1-15	“There’s a disconnect between words and lessons... words from the lessons... aren’t appearing here in the words section.”	Words and lessons relationship is conceptually unclear	Transcript 2, Speaker 1 (22:07–22:45)
T2-S1-16	“The pin is here if I was to send an email.”	PIN recovery raises visibility/privacy concern	Transcript 2, Speaker 1 (21:27–21:49)
T2-S1-17	“I’d be concerned that this wouldn’t be giving them... to take their time with pronouncing these words.”	Time pressure may conflict with careful pronunciation practice	Transcript 2, Speaker 1 (27:06–27:46)
T2-S1-18	“The feedback gives them something positive... because there’s another enemy that then appears, and then is killed.”	Delayed recognition may produce misleading positive feedback	Transcript 2, Speaker 1 (28:00–28:57)
T2-S1-19	“A global leaderboard... would perhaps be quite intimidating for a child... a local leaderboard... would probably be better.”	Global comparison may intimidate; local comparison may be safer	Transcript 2, Speaker 1 (29:59–30:37)
T2-S1-20	“Dashboard with the custom words... tailoring therapy to a child... generates a report.”	Dashboard enables tailored therapy and reportable progress	Transcript 2, Speaker 1 (30:54–31:25)
T2-S1-21	“I like the controls... moving up-and-down... it’s more of a gameplay element.”	Movement controls create active gameplay beyond repetition	Transcript 2, Speaker 1 (31:25–31:48)
T2-S1-22	“Landscape mode was a lot more beneficial... giving more information.”	Landscape presents the game state more honestly and usefully	Transcript 2, Speaker 1 (32:29–33:07)